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Existence, uniqueness and comparison theorem on unbounded solutions of scalar super-linear BSDEs[☆]

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Abstract

This paper is devoted to the existence, uniqueness and comparison theorem on unbounded solutions of a scalar backward stochastic differential equation (BSDE) whose generator grows (with respect to both unknown variables y and z) in a super-linear way like $|y||\ln|y||^{\delta} + |z||\ln|z||^{\lambda}$ for some $\delta \in [0, 1]$ and $\lambda \geq 0$. Let k be the maximum of δ, $\lambda + 1/2$ and 2λ. For the following four different ranges of the growth power parameter k: $k = 1/2$, $k \in (1/2, 1)$, $k = 1$ and $k > 1$, we give reasonably weakest possible different integrability conditions on the terminal value for ensuring existence and uniqueness of the unbounded solution to the BSDE. In the first two cases, they are stronger than the $L \ln L$-integrability and weaker than any L^p-integrability with $p > 1$; in the third case, the integrability condition is just some L^p-integrability for $p > 1$; and in the last case, the integrability condition is stronger than any L^p-integrability with $p > 1$ and weaker than any $\exp(L^{\varepsilon})$-integrability with $\varepsilon \in (0, 1)$. We also establish three comparison theorems, which yield immediately the uniqueness, when either one of generators of both BSDEs is convex (or concave) in both unknown variables (y, z), or satisfies a one-sided Osgood condition in the first unknown variable y and a uniform continuity condition in the second unknown variable z. Finally, we give an application of our results in mathematical finance.

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1. Introduction

We fix a positive real number $T > 0$ and a positive integer d, and let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space, $(B_t)_{t\in[0,T]}$ an $\mathbb{R}^d$-valued standard Brownian motion defined on this space, and $(\mathcal{F}_t)_{t\in[0,T]}$ the natural filtration generated by B and augmented by all $\mathbb{P}$-null sets of $\mathcal{F}$. Any progressive measurability and adaptability of processes will be associated to this filtration.

Let $\mathbb{R}_+$ be the set of all nonnegative real numbers, $x\cdot y$ the scalar inner product of two vectors $x, y \in \mathbb{R}^d$, $\mathbf{1}_A$ the indicator function of set A, and $\mathrm{sgn}(x) := \mathbf{1}_{x>0} - \mathbf{1}_{x\le 0}$. We recall that a real-valued and progressively measurable process $(X_t)_{t\in[0,T]}$ belongs to class (D) if the family of random variables $\{X_\tau : \tau$ is any $(\mathcal{F}_t)$-stopping time taking values in $[0, T]\}$ is uniformly integrable.

Consider the following backward stochastic differential equation (BSDE for short):

$$Y_t = \xi + \int_t^T g(s, Y_s, Z_s)\mathrm{d}s - \int_t^T Z_s \cdot \mathrm{d}B_s, \quad t \in [0, T], \tag{1.1}$$

where the terminal condition ξ is a real-valued $\mathcal{F}_T$-measurable random variable, and the generator

$$g(\omega, t, y, z) : \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}$$

is jointly continuous in (y, z) and adapted for each (y, z). A solution of (1.1) is a pair of adapted processes $(Y_t, Z_t)_{t\in[0,T]}$ with values in $\mathbb{R} \times \mathbb{R}^d$, such that $\mathbb{P}$-*a.s.*, the functions $t \mapsto Y_t$ is continuous, $t \mapsto |Z_t|^2 + |g(t, Y_t, Z_t)|$ is integrable, and (1.1) is satisfied. We denote by BSDE(ξ, g) the BSDE with the terminal value ξ and the generator g. All definitions of functional spaces used later will be placed at the end of this section.

In this paper, let us always assume that $(\alpha_t)_{t\in[0,T]}$ is an $\mathbb{R}_+$-valued adapted integrable process, and that $\beta \ge 0$, $\gamma > 0$, $\delta \in [0, 1]$, $\lambda \ge 0$ and $c > 0$ are given nonnegative constants. We consider the unbounded solution of BSDE (1.1) with a one-sided super-linearly growing generator g, which satisfies, roughly speaking, that $\mathrm{d}\mathbb{P} \times \mathrm{d}t$-*a.e.*,

$$\forall (y, z) \in \mathbb{R} \times \mathbb{R}^d, \quad \mathrm{sgn}(y)g(\omega, t, y, z) \le \alpha_t(\omega) + \beta|y|(\ln|y|)^\delta \mathbf{1}_{|y|>1} + \gamma|z|| \ln|z||^\lambda. \tag{1.2}$$

It is clear that the second term in the right-hand side of (1.2) can be replaced equivalently with the term $\beta|y|| \ln|y||^\delta$ since the absolute value of their difference is less than $1/e$. And, note that for each $\delta > 1$,

$$\int_0^{+\infty} \frac{1}{(y+e)(\ln(y+e))^\delta}\mathrm{d}y = \frac{1}{\delta - 1} =: \bar{\delta} < +\infty.$$

When $\delta > 1$ and $T > 2\bar{\delta}$, by separating variables, we see that the following ODE

$$y'(t) = (|y(t)| + e)(\ln(|y(t)| + e))^\delta, \quad t \in [0, T] \tag{1.3}$$

has no global solution on $[0, T]$ with the terminal value a. However, for $\delta \in [0, 1]$, it is easily verified that the preceding ODE (1.3) admits indeed a unique global solution on $[0, T]$ with a given terminal value a. Since an ODE is a special BSDE with the second unknown variable z being vanishing, we only consider $\delta \in [0, 1]$ in (1.2). The reader is referred to [1] for the existence or nonexistence results on BSDEs in the case of $\delta > 1$.

The generator g is said to be of linear growth when $\delta = \lambda = 0$ and the left-hand side of (1.2) is replaced with $|g(\omega, t, y, z)|$. In this case, if the terminal condition $(\xi, \alpha_\cdot)$ belongs to $L^p \times \mathcal{L}^p$ for some $p > 1$, then BSDE(ξ, g) admits a solution in the space $\mathcal{S}^p \times \mathcal{M}^p$, and the solution is unique in this space if g further satisfies the uniformly Lipschitz condition or certain uniform continuity condition in (y, z). The reader is referred to [2–9] for details. Recently, the authors in [10–12] (see also [13]) found step by step a weakest integrability condition for $(\xi, \alpha_\cdot)$ to guarantee the existence of a solution to a linearly growing BSDE(ξ, g), where the terminal condition $(\xi, \alpha_\cdot)$ only needs to satisfy an $L \exp\left(\mu\sqrt{\ln(1 + L)}\right)$-integrability condition for a positive parameter $\mu = \gamma\sqrt{2T}$. They showed that this integrability condition is weaker than the usual L^p-integrability with $p > 1$ and stronger than the usual $L \ln L$-integrability, and that the preceding integrability for a positive parameter $\mu < \gamma\sqrt{2T}$ is not sufficient to ensure the existence of a solution. They also established the uniqueness of the unbounded solution under the conditions that g satisfies the monotonicity condition in y and the uniformly Lipschitz condition in z. Generally speaking, the generator g can allow a general growth in the first unknown variable y when g satisfies the (weak) monotonicity condition in y. Related works can be found in [14–23]. However, we would like to point out that the (weak) monotonicity condition in y implies that g has a one-sided linear growth in y, see [24] for details.

Furthermore, Bahlali et al. [25,26] proved the existence of a solution to BSDE(ξ, g) in the space $\mathcal{S}^p \times \mathcal{M}^2$ for some sufficiently large $p > 2$, when the terminal condition $(\xi, \alpha_\cdot)$ belongs to $L^p \times \mathcal{L}^p$ and the generator g satisfies (1.2) with $\delta = 1$, $\lambda = 1/2$ and $|g(\omega, t, y, z)|$ instead of the left side of (1.2). They also established the uniqueness of the solution when g further satisfies a locally monotone condition in both unknown variables (y, z). Some related works on BSDEs with super-linearly growing generators are available in [27–34], where the solution of BSDE(ξ, g) in the space $\mathcal{S}^p \times \mathcal{M}^p$ is considered under the assumption that the terminal condition $(\xi, \alpha_\cdot) \in L^p \times \mathcal{L}^p$ for some $p > 1$, and some kinds of locally Lipschitz conditions or locally monotone conditions of g in (y, z) are usually used in order to ensure the uniqueness of the solution of BSDE(ξ, g).

Finally, for quadratic BSDEs, namely, the generator g has a quadratic growth in the second unknown variable z, much progress has been made in the last two decades. Roughly speaking, to have the existence of a solution to BSDE(ξ, g), the boundedness or at least some exponential integrability of the terminal condition $(\xi, \alpha_\cdot)$ are required. The reader is referred to [22,35–43] among others. We also mention that a class of quadratic BSDEs with L^p-integrable terminal conditions with $p > 1$ are investigated in several recent works, see for example [44–47]. For super-quadratic BSDEs, some intensive results can be found in for example [48–52]. Very recently, well-posedness of scalar BSDEs with sub-quadratic generators was investigated in [53], where the terminal condition $(\xi, \alpha_\cdot)$ needs to satisfy an $\exp(\mu L^r)$-integrability condition for some $\mu > 0$ and $r \in (0, 1)$ weaker than the $\exp(\mu L)$-integrability and stronger than any L^p-integrability with $p > 1$. In particular, we would like to mention that when the growth of the generator g in both unknown variables (y, z) is not limited to be linear, convexity or concavity of g in (y, z) are usually required in order to ensure the uniqueness of the solution of BSDE(ξ, g), as has been done in for example [5,22,37,38,45,48,53–56].

Enlightened by these results mentioned above, this paper is devoted to the existence, uniqueness and comparison theorem for unbounded solutions of BSDEs with generator g having a super-linear growth as in (1.2). Let $k := \delta \vee (\lambda + 1/2) \vee (2\lambda)$, which is the maximum of δ, $\lambda + 1/2$ and 2λ. We distinguish four different situations: (i) $k = 1/2$, (ii) $k \in (1/2, 1)$, (iii) $k = 1$, (iv) $k > 1$, and put forward four reasonably weakest possible integrability conditions on the terminal value $(\xi, \alpha_\cdot)$ to ensure existence and uniqueness of the unbounded solution to the BSDE(ξ, g), see Theorem 2.4 in Section 2 for details. In the first two cases, they are stronger than the $L \ln L$-integrability and weaker than any L^p-integrability with $p > 1$; in the third case, the integrability condition is just some L^p-integrability for $p > 1$; and in the last case, the integrability condition is stronger than any L^p-integrability with $p > 1$ and weaker than any $\exp(L^\varepsilon)$-integrability with $\varepsilon \in (0, 1)$. In particular, Theorem 2.4 explores the roles of each of parameters $(\beta, \gamma, \delta, \lambda)$ appearing in (1.2) played in the weakest possible integrability conditions on $(\xi, \alpha_\cdot)$, which is the main novelty of this paper, see Remark 2.5 in Section 2 for more details. Several examples (see Examples 2.6–2.9) are also provided in Section 2 to illustrate applications of Theorem 2.4. Furthermore, we establish three comparison theorems of unbounded solutions for the BSDEs (see Theorems 2.11–2.13 in Section 2), which yield naturally the uniqueness, when either one of generators of both BSDEs is convex (concave) in both unknown variables (y, z), or satisfies a one-sided Osgood condition in the first unknown variable y and a uniform continuity condition in the second unknown variable z. They improve those related results of [5,11–13], and see Remark 2.14 in Section 2 for further details. All these results of the paper seem to be new. They are fundamental for super-linear growing BSDEs, and pave a way to their practical applications in vary mathematical fields including partial differential equations, stochastic control, stochastic games, mathematical finance and others.

We end this section by introducing some notations of functional spaces. For any real $p \geq 1$, L^p is the set of all real-valued and $\mathcal{F}_T$-measurable random variables ξ satisfying $\mathbb{E}[|\xi|^p] < +\infty$, $\mathcal{L}^p$ is the set of all real-valued adapted processes $(X_t)_{t\in[0,T]}$ such that

$$\|X\|_{\mathcal{L}^p} := \left\{ \mathbb{E}\left[\left(\int_0^T |X_t| \mathrm{d}t \right)^p \right] \right\}^{1/p} < +\infty,$$

$\mathcal{S}^p$ is the set of all real-valued adapted continuous processes $(Y_t)_{t\in[0,T]}$ satisfying

$$\|Y\|_{\mathcal{S}^p} := \left(\mathbb{E}[\sup_{t\in[0,T]} |Y_t|^p] \right)^{1/p} < +\infty,$$

and $\mathcal{M}^p$ is the set of all $\mathbb{R}^d$-valued adapted processes $(Z_t)_{t\in[0,T]}$ such that

$$\|Z\|_{\mathcal{M}^p} := \left\{ \mathbb{E}\left[\left(\int_0^T |Z_t|^2 \mathrm{d}t \right)^{p/2} \right] \right\}^{1/p} < +\infty.$$

The rest of this paper is organized as follows. In Section 2, we state the existence, uniqueness and comparison theorems for unbounded solutions of BSDEs together with an application in mathematical finance by five subsections. These results are proved in Section 3. For ease of presentation, the proofs of several technical propositions are given in the Appendix.

2. Main results

We always suppose that ξ is a terminal condition and g is a generator which is jointly continuous in both unknown variables (y, z).

2.1. A key function and several ODEs

For convenience of subsequent exposition, we introduce the constant

$$k := \delta \vee (\lambda + 1/2) \vee (2\lambda) \in [1/2, +\infty) \tag{2.1}$$

and the function

$$\psi(x, \mu; k) := x \exp\left(\mu\, (\ln(1+x))^k\right), \quad (x, \mu) \in \mathbb{R}_+ \times \mathbb{R}_+. \tag{2.2}$$

It can be verified that

(i) if $k \in [1/2, 1)$, then $k = \delta \vee (\lambda + 1/2)$, and for each $p > 1$,

$$x \ln(1+x) < \psi(x, \mu; k) = x \exp\left(\mu\, (\ln(1+x))^{\delta\vee(\lambda+1/2)}\right) < x^p, \quad (x, \mu) \in \mathbb{R}_+ \times (0, +\infty); \tag{2.3}$$

(ii) if $k = 1$, then $\delta = 1$ or $\lambda = 1/2$, and

$$x^{1+\mu} < \psi(x, \mu; k) = x(1+x)^\mu < (1+x)^{1+\mu}, \quad (x, \mu) \in \mathbb{R}_+ \times (0, +\infty); \tag{2.4}$$

(iii) if $k \in (1, +\infty)$, then $k = 2\lambda$, and for each $p > 1$ and $\epsilon \in (0, 1)$,

$$x^p < \psi(x, \mu; k) = x \exp\left(\mu\, (\ln(1+x))^{2\lambda}\right) < \exp(x^\epsilon), \quad (x, \mu) \in \mathbb{R}_+ \times (0, +\infty). \tag{2.5}$$

For each $\varepsilon > 0$, denote

$$\beta_{k,\varepsilon} = \begin{cases} \beta \mathbf{1}_{\delta=1/2} + \beta\varepsilon \mathbf{1}_{0\le\delta<1/2}, & k = 1/2; \\ \beta \mathbf{1}_{\delta\ge\lambda+1/2} + \beta\varepsilon \mathbf{1}_{0\le\delta<\lambda+1/2}, & k = (1/2, 1); \\ \beta \mathbf{1}_{\delta=1} + \beta\varepsilon \mathbf{1}_{0\le\delta<1}, & k \ge 1 \end{cases}$$

and

$$\gamma_{k,\varepsilon} = \begin{cases} \gamma, & k = 1/2; \\ \gamma \mathbf{1}_{\lambda\ge\delta-1/2} + \gamma\sqrt{\varepsilon} \mathbf{1}_{0\le\lambda<\delta-1/2}, & k = (1/2, 1); \\ \gamma \mathbf{1}_{\lambda=1/2} + \gamma\sqrt{\varepsilon} \mathbf{1}_{0\le\lambda<1/2}, & k = 1; \\ \gamma, & k > 1. \end{cases}$$

It is clear that as $\varepsilon \to 0^+$, $(\beta_{k,\varepsilon}, \gamma_{k,\varepsilon})$ tends decreasingly to the following $(\beta_{k,0}, \gamma_{k,0})$ with

$$\beta_{k,0} = \begin{cases} \beta \mathbf{1}_{\delta=1/2}, & k = 1/2; \\ \beta \mathbf{1}_{\delta\ge\lambda+1/2}, & k = (1/2, 1); \\ \beta \mathbf{1}_{\delta=1}, & k \ge 1 \end{cases} \quad \textit{and} \quad \gamma_{k,0} = \begin{cases} \gamma, & k = 1/2; \\ \gamma \mathbf{1}_{\lambda\ge\delta-1/2}, & k = (1/2, 1); \\ \gamma \mathbf{1}_{\lambda=1/2}, & k = 1; \\ \gamma, & k > 1. \end{cases}$$

Furthermore, for $\varepsilon > 0$, let $\mu_{k,\varepsilon}(\cdot)$ be the unique solution of the following ODE: $\mu_{k,\varepsilon}(0) = \varepsilon$ and

$$\mu'_{k,\varepsilon}(s) = \begin{cases} \dfrac{\gamma_{k,\varepsilon}^2}{\mu_{k,\varepsilon}(s)} + \beta_{k,\varepsilon}, & k = 1/2; \\ \beta_{k,\varepsilon}\left(\lambda + \dfrac{1}{2}\right)\mu_{k,\varepsilon}(s) + \dfrac{\gamma_{k,\varepsilon}^2(1+\varepsilon)^{2k+1}}{2k}\dfrac{1}{\mu_{k,\varepsilon}(s)} + \beta_{k,\varepsilon} + \varepsilon, & k \in (1/2, 1); \\ \beta_{k,\varepsilon}(\mu_{k,\varepsilon}(s) + 1) + \dfrac{\gamma_{k,\varepsilon}^2(1+\varepsilon)}{2}\left(\dfrac{1}{\mu_{k,\varepsilon}(s)} + 1\right), & k = 1; \\ k\beta_{k,\varepsilon}\mu_{k,\varepsilon}(s) + \dfrac{\gamma_{k,\varepsilon}^2(1+\varepsilon)\bar{k}_\lambda}{2}\left(1 + \dfrac{\varepsilon}{k}\mu_{k,\varepsilon}(s)\right) + \beta_{k,\varepsilon}\varepsilon, & k > 1 \end{cases} \quad \text{for } s \in [0, T], \tag{2.6}$$

and $\mu_{k,0}(\cdot)$ be the unique solution of the following ODE: $\mu_{k,0}(0) = 0$ and

$$\mu'_{k,0}(s) = \begin{cases} \dfrac{\gamma_{k,0}^2}{2k}\dfrac{1}{\mu_{k,0}(s)} + \beta_{k,0}, & k \in [1/2, 1); \\ \beta_{k,0}(\mu_{k,0}(s) + 1) + \dfrac{\gamma_{k,0}^2}{2}\left(\dfrac{1}{\mu_{k,0}(s)} + 1\right), & k = 1; \\ k\beta_{k,0}\mu_{k,0}(s) + \dfrac{\gamma_{k,0}^2}{2}\bar{k}_\lambda, & k > 1 \end{cases} \quad \text{for } s \in (0, T] \tag{2.7}$$

with

$$\bar{k}_\lambda := 2^{2(\lambda-1)^+ + 2\lambda - 1}. \tag{2.8}$$

It is not difficult to check that as $\varepsilon \to 0^+$, $\mu_{k,\varepsilon}(\cdot)$ tends decreasingly to $\mu_{k,0}(\cdot)$ on $[0, T]$.

Remark 2.1. With regard to the solution of the ODE in (2.7), we make the following three remarks.

(i) By separating variables we can solve the ODE in (2.7) when $k \in [1/2, +\infty)$ and $k \neq 1$. Indeed, for $k \in [1/2, 1)$ and $t \in [0, T]$ we have

$$\mu_{k,0}(t) = \begin{cases} \gamma_{k,0}\sqrt{\dfrac{t}{k}}, & \beta_{k,0} = 0; \\ \beta_{k,0}t, & \beta_{k,0} > 0 \ \text{ and } \ \gamma_{k,0} = 0; \\ \dfrac{\gamma_{k,0}^2}{2k\beta_{k,0}}\ln\left(1 + \dfrac{2k\beta_{k,0}}{\gamma_{k,0}^2}\mu_{k,0}(t)\right) - \beta_{k,0}t, & \beta_{k,0} > 0 \ \text{ and } \ \gamma_{k,0} > 0, \end{cases} \tag{2.9}$$

and for $k \in (1, +\infty)$ and $t \in [0, T]$ we have

$$\mu_{k,0}(t) = \begin{cases} \dfrac{\gamma_{k,0}^2}{2}\bar{k}_\lambda t, & \beta_{k,0} = 0; \\ \dfrac{\gamma_{k,0}^2}{2k\beta_{k,0}}\bar{k}_\lambda\left(e^{k\beta_{k,0}t} - 1\right), & \beta_{k,0} > 0. \end{cases} \tag{2.10}$$

(ii) For the case of $\beta_{k,0} > 0$ and $\gamma_{k,0} > 0$ in (2.9), by virtue of the Lambert function (see [57] for details) we can get the following explicit expression of $\mu_{k,0}(\cdot)$:

$$\mu_{k,0}(t) = -\frac{\gamma_{k,0}^2}{2k\beta_{k,0}}\left[W\left(-e^{-\frac{2k\beta_{k,0}^2}{\gamma_{k,0}^2}t-1}\right)+1\right], \quad t \in [0, T], \tag{2.11}$$

where $W(\cdot)$ is the Lambert function defined on $[-1/e, 0)$ satisfying

$$W(x)e^{W(x)} = x \quad \text{and} \quad W(x) \le -1.$$

Furthermore, utilizing Theorem 1 in [57] we can get the following upper bound of $\mu_{k,0}(\cdot)$:

$$\mu_{k,0}(t) \le -\frac{\gamma_{k,0}^2}{2k\beta_{k,0}}\left(-\sqrt{\frac{4k\beta_{k,0}^2}{\gamma_{k,0}^2}t} - \frac{2k\beta_{k,0}^2}{\gamma_{k,0}^2}t\right) = \gamma_{k,0}\sqrt{\frac{t}{k}} + \beta_{k,0}t, \quad t \in [0, T]. \tag{2.12}$$

(iii) For the case of $k = 1$ in (2.7), by separating variables we can also get an implicit expression of the function $\mu_{k,0}(\cdot)$. But, it seems to be difficult to get its explicit expression. However, we can find its upper bound by virtue of the comparison theorem of ODEs. In fact, in the case of $\beta_{k,0} = 0$ and $\gamma_{k,0} > 0$, a same computation as in (ii) yields that

$$\mu_{k,0}(t) \le \gamma_{k,0}\sqrt{t} + \frac{\gamma_{k,0}^2}{2}t, \quad t \in [0, T]. \tag{2.13}$$

Now we suppose that $\beta_{k,0} > 0$. For each $\varepsilon > 0$, let $\bar{\mu}_\varepsilon(\cdot)$ be the unique solution of the following ODE:

$$\bar{\mu}_\varepsilon(0) = \varepsilon \quad \text{and} \quad \bar{\mu}'_\varepsilon(s) = \frac{\gamma_{k,0}^2}{2}\frac{1}{\bar{\mu}_\varepsilon(s)} + \beta_{k,0}\bar{\mu}_\varepsilon(s) + \beta_{k,0} + \frac{\gamma_{k,0}^2}{2}, \quad t \in [0, T]. \tag{2.14}$$

It is clear that $\mu_{k,0}(t) \le \bar{\mu}_\varepsilon(t)$ for each $t \in [0, T]$ and $\varepsilon > 0$. On the other hand, by (2.14) we have

$$\bar{\mu}'_\varepsilon(s) \le \beta_{k,0}\bar{\mu}_\varepsilon(s) + \beta_{k,0} + \frac{\gamma_{k,0}^2}{2}(1 + \frac{1}{\varepsilon}), \quad t \in [0, T],$$

and then

$$\bar{\mu}_\varepsilon(t) \le \varepsilon + \beta_{k,0}t + \frac{\gamma_{k,0}^2}{2}(1 + \frac{1}{\varepsilon})t + \beta_{k,0}\int_0^t \bar{\mu}_\varepsilon(s)\mathrm{d}s, \quad t \in [0, T].$$

It then follows from Gronwall's inequality that for each $\varepsilon > 0$,

$$\bar{\mu}_\varepsilon(t) \le \left[\varepsilon + \frac{\gamma_{k,0}^2 t}{2\varepsilon} + \left(\beta_{k,0} + \frac{\gamma_{k,0}^2}{2}\right)t\right]e^{\beta_{k,0}t}, \quad t \in [0, T].$$

Hence, by picking $\varepsilon = \gamma_{k,0}\sqrt{t/2}$ we obtain that

$$\mu_{k,0}(t) \le \bar{\mu}_\varepsilon(t) \le \left[\gamma_{k,0}\sqrt{2t} + \left(\beta_{k,0} + \frac{\gamma_{k,0}^2}{2}\right)t\right]e^{\beta_{k,0}t}, \quad t \in [0, T]. \tag{2.15}$$

2.2. *Basic assumptions*

For the existence of a solution of BSDE(ξ, g), we need the following assumptions (EX1) and (EX2) on the generator g.

(EX1) g has a general growth in y and a quadratic growth in z, i.e., there exists an $\mathbb{R}_+$-valued adapted integrable process $(f_t)_{t\in[0,T]}$, and a nonnegative real-valued function $H(\cdot)$ defined on $\mathbb{R}_+$ with $H(0)=0$ such that $\mathrm{d}\mathbb{P}\times\mathrm{d}t$-$a.e.$,

$$\forall (y,z)\in\mathbb{R}\times\mathbb{R}^d,\quad |g(\omega,t,y,z)|\leq f_t(\omega)+H(|y|)+c|z|^2.$$

(EX2) g has a one-sided super-linear growth in (y,z), i.e., $\mathrm{d}\mathbb{P}\times\mathrm{d}t$-$a.e.$,

$$\forall (y,z)\in\mathbb{R}\times\mathbb{R}^d,\quad \mathrm{sgn}(y)g(\omega,t,y,z)\leq\alpha_t(\omega)+\beta|y|(\ln|y|)^{\delta}\mathbf{1}_{|y|>1}+\gamma|z|\|\ln|z|\|^{\lambda}.$$

For the uniqueness of the solution of BSDE(ξ,g), we need the following assumptions (UN1) and (UN2) or (UN3) on the generator g.

(UN1) g satisfies a one-sided Osgood condition in y, i.e., there exists a nonnegative, increasing, continuous and concave function $\rho(\cdot)$ defined on $\mathbb{R}_+$ satisfying $\rho(0)=0$, $\rho(u)>0$ for $u>0$ and

$$\int_{0^+}\frac{\mathrm{d}u}{\rho(u)}:=\lim_{\varepsilon\to 0^+}\int_0^{\varepsilon}\frac{\mathrm{d}u}{\rho(u)}=+\infty$$

such that $\mathrm{d}\mathbb{P}\times\mathrm{d}t$-$a.e.$,

$$\forall (y_1,y_2,z)\in\mathbb{R}\times\mathbb{R}\times\mathbb{R}^d,\quad \mathrm{sgn}(y_1-y_2)\left(g(\omega,t,y_1,z)-g(\omega,t,y_2,z)\right)\leq\rho(|y_1-y_2|).$$

(UN2) g is uniformly continuous in z, i.e., there exists a nonnegative, nondecreasing and continuous function $\kappa(\cdot)$ defined on $\mathbb{R}_+$ with $\kappa(0)=0$ such that $\mathrm{d}\mathbb{P}\times\mathrm{d}t$-$a.e.$,

$$\forall (y,z_1,z_2)\in\mathbb{R}\times\mathbb{R}^d\times\mathbb{R}^d,\quad |g(\omega,t,y,z_1)-g(\omega,t,y,z_2)|\leq\kappa(|z_1-z_2|).$$

(UN3) $\mathrm{d}\mathbb{P}\times\mathrm{d}t$-$a.e.$, g is convex or concave in (y,z).

Remark 2.2. Since the function $\rho:\mathbb{R}_+\to\mathbb{R}_+$ defined in (UN1) is nondecreasing and concave with $\rho(0)=0$, it follows from Lemma 6.1 in [19] that $\rho(x)/x$ $(x>0)$ is non-increasing and then $\rho(\cdot)$ has a linear growth. And, in (UN2), since g is uniformly continuous in z, the continuous modular function $\kappa(\cdot)$ can be chosen to have a linear growth. Consequently, without loss of generality, we always assume that the functions $\rho(\cdot)$ and $\kappa(\cdot)$ has a linear growth, i.e., there exists a positive constant $A>0$ such that

$$\forall u\in\mathbb{R}_+,\quad \rho(u)\leq A(u+1)\ \text{ and }\ \kappa(u)\leq A(u+1).$$

Remark 2.3. Assumptions (UN1) and (UN2) are strictly weaker than the usual monotonicity of the generator g in y and the uniformly Lipschitz continuity of g in z, respectively. They are usually used when the generator g has a general growth in y and a linear growth in z. See [19,20,22] among others for more details. The generator g is also assumed to be convex (or concave) in z for the uniqueness of solutions of BSDEs in for example [37,38]. In particular, assumptions (UN1) and (UN2) imply assumption (EX2) with $(\delta,\lambda)=(0,0)$, which is the special case of $k=1/2$.

2.3. Well-posedness of BSDEs

The following existence and uniqueness theorem is the main result of this paper.

Theorem 2.4. *Let the generator g satisfy assumptions (EX1) and (EX2), the constant k be defined in (2.1), and the functions $\psi(\cdot,\cdot;k)$, $\mu_{k,\varepsilon}(\cdot)$ and $\mu_{k,0}(\cdot)$ be respectively defined in (2.2),* (2.6) and (2.7). *If there exists a constant* $\mu > \mu_{k,0}(T)$ *(resp.* $\mu \geq \mu_{k,0}(T)$ *in the case of* $k = \delta = 1/2$*) such that*

$$\mathbb{E}\left[\psi\left(|\xi| + \int_0^T \alpha_t \mathrm{d}t,\ \mu;\ k\right)\right] < +\infty, \tag{2.16}$$

then BSDE(ξ, g) *has a solution* $(Y_t, Z_t)_{t\in[0,T]}$ *such that* $\psi\left(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, \mu_{k,\varepsilon}(\cdot); k\right)$ *belongs to class (D), where* $\varepsilon > 0$ *(resp.* $\varepsilon \geq 0$*) is the unique constant satisfying* $\mu_{k,\varepsilon}(T) = \mu$*. And, there exists a positive constant* $C > 0$ *depending only on* $(k, \beta, \gamma, \lambda, \varepsilon, T)$ *such that* $\mathbb{P}$*-a.s., for each* $t \in [0, T]$*,*

$$|Y_t| \leq \psi\left(|Y_t| + \int_0^t \alpha_s \mathrm{d}s,\ \mu_{k,\varepsilon}(t);\ k\right) \leq C\mathbb{E}\left[\psi\left(|\xi| + \int_0^T \alpha_s \mathrm{d}s,\ \mu;\ k\right)\middle|\mathcal{F}_t\right] + C. \tag{2.17}$$

Moreover, we have

(i) If $k = 1/2$*, then* $(Y_\cdot, Z_\cdot) \in \mathcal{S}^p \times \mathcal{M}^p$ *for each* $p \in (0, 1)$*. And, if* g *also satisfies either assumptions* (UN1) *and* (UN2) *or* (UN3)*, then the solution* $(Y_\cdot, Z_\cdot)$ *such that* $\psi\left(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, \mu_{k,\varepsilon}(\cdot); k\right)$ *belongs to class (D) for some* $\varepsilon > 0$ *(resp.* $\varepsilon \geq 0$ *in case of* $k = \delta = 1/2$*) is unique.*

(ii) If $k \in (1/2, 1)$*, then* $(Y_\cdot, Z_\cdot) \in \mathcal{S}^p \times \mathcal{M}^p$ *for each* $p \in (0, 1)$*. And, if* g *also satisfies* (UN3)*, then the solution* $(Y_\cdot, Z_\cdot)$ *such that the process* $\psi\left(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, \mu_{k,\varepsilon}(\cdot); k\right)$ *belongs to class (D) for some* $\varepsilon > 0$ *is unique.*

(iii) If $k = 1$*, then* $(Y_\cdot, Z_\cdot) \in \mathcal{S}^p \times \mathcal{M}^p$ *for each* $p \in [1, 1 + \varepsilon)$*. And, if the generator* $g = g_1 + g_2$*,* g_1 *satisfies assumptions* (UN1) *and* (UN2) *and* g_2 *satisfies assumption* (UN3)*, then the solution* $(Y_\cdot, Z_\cdot)$ *such that* (2.17) *holds for each* $t \in [0, T]$ *and some* $\varepsilon > 0$ *is unique.*

(iv) If $k > 1$*, then* $(Y_\cdot, Z_\cdot) \in \mathcal{S}^p \times \mathcal{M}^p$ *for each* $p \geq 1$*. And, if the condition* (2.16) *holds for some* $\mu > 2q3^{k-1}\mu_{k,0}(T)$ *with* $q \geq 1$*, then BSDE*(ξ, g) *admits a solution* $(Y_t, Z_t)_{t\in[0,T]}$ *such that the process* $\psi\left(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s,\ 2q3^{k-1}\mu_{k,\varepsilon}(\cdot); k\right)$ *belongs to class (D), where* $\varepsilon > 0$ *is the unique positive constant satisfying* $2q3^{k-1}\mu_{k,\varepsilon}(T) = \mu$*. If* g *further satisfies* (UN3)*, then the solution such that the process* $\psi\left(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s,\ 2q3^{k-1}\mu_{k,\varepsilon}(\cdot); k\right)$ *belongs to class (D) for some* $\varepsilon > 0$ *is also unique.*

The integrability condition (2.16) seems to be reasonably weakest possible for existence and uniqueness of the solution of BSDE(ξ, g). It would be interesting to prove or disprove it.

Remark 2.5. Concerning Theorem 2.4, we have the following comments.

(i) In (EX2), the super-linear growth of the generator g in (y, z) is featured by parameters $(\alpha_\cdot, \beta, \gamma, \delta, \lambda)$, all of which enter into (2.6), (2.7) and (2.16). Theorem 2.4 shows that the integrability of data (g, ξ), required for the existence and uniqueness result on BSDE(ξ, g), and then the integrability of the solution $(Y_\cdot, Z_\cdot)$ of this BSDE, increase with anyone of the five parameters. In particular, in view of (2.3)–(2.5), the integrability changes in a distinct way among the three cases: $k \in [1/2, 1)$, $k = 1$ and $k > 1$. In addition, for the uniqueness of the solution, if only the assumptions on the generator g are considered, they are weakest in the case of $k = 1$.

(ii) When $k = 1/2$, λ has to be zero and δ can take values in the interval $[0, 1/2]$. The case of $\lambda = \delta = 0$ has been discussed in [12], where some stronger assumptions than (UN1) and (UN2) on the generator g are required for the uniqueness of the solution of BSDE(ξ, g). However, by an exponential shift transform, it has been shown in [12] that μ can be equal to

$\mu_{k,0}(T) = \gamma\sqrt{2T}$ when $\lambda = \delta = 0$, which is better than our conditions in this case. In fact, except the two cases that $\delta = \lambda = 0$ and $k = \delta = 1/2$, μ in (2.16) is required to be strictly bigger than $\mu_{k,0}(T)$. In addition, from the definitions of $\beta_{k,0}$ and $\gamma_{k,0}$, the range of μ changes abruptly at several critical points (for example $\delta = 1/2$, $\delta = 1$, $\lambda = 1/2$ and $\delta = \lambda + 1/2$) for the existence and uniqueness result on BSDE(ξ, g).

(iii) In the case of $\delta = 1$ and $\lambda = 1/2$ (a special case of $k = 1$), the well-posedness of BSDEs has been given in [25,26]. In comparison with their related results, weaker assumptions on both the terminal condition $(\xi, \alpha_\cdot)$ and the generator g are required in Theorem 2.4 for the existence of solutions of BSDE(ξ, g). In fact, in Theorem 2.4, $(\xi, \alpha_\cdot)$ may only need to belong to L^p for some $1 < p \le 2$, whereas $p > 2$ in [25,26], and g only needs to satisfy (1.2) with $\delta = 1$ and $\lambda = 1/2$, whereas the left side of (1.2) needs to be replaced with $|g(\omega, t, y, z)|$ in [25,26]. In addition, to ensure the uniqueness of the solution of BSDE(ξ, g), a locally monotone condition is required in [25,26], which is different from the assumptions in (iii) of Theorem 2.4.

In the sequel, we introduce four examples which Theorem 2.4 can apply to, but any existing results cannot apply to. They correspond to four different cases of the range of k respectively, that is, $k = 1/2$, $k \in (1/2, 1)$, $k = 1$ and $k > 1$.

Example 2.6. Define for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$,

$$g_1(\omega, t, y, z) := y - e^y + h(|y|) + \left(|z| - \sqrt{|z|}\right) + 2\sin|z|$$

and

$$g_2(\omega, t, y, z) := |B_t(\omega)| + e^{-y} + |y|\sqrt[3]{\ln|y|}\mathbf{1}_{|y|\ge 1} + |z|,$$

where

$$h(x) := x\sqrt{|\ln x|}\mathbf{1}_{0\le x\le\varepsilon} + h'_-(\varepsilon)(x - h(\varepsilon)), \quad x \in \mathbb{R}_+$$

with $\varepsilon > 0$ being sufficiently small. Then, the generator g_1 satisfies assumptions (EX1) with $c = 2$, (EX2) with $\alpha_\cdot \equiv 5, \beta = 1, \gamma = 2, \delta = 0, \lambda = 0$, (UN1) with $\rho(x) = x + h(x)$ and (UN2) with $\kappa(x) = 3x + \sqrt{x}$, and the generator g_2 satisfies assumptions (EX1) with $c = 1$, (EX2) with $\alpha_\cdot = |B_\cdot| + 1, \beta = 1, \gamma = 1, \delta = 1/3, \lambda = 0$ and (UN3). Hence, both g_1 and g_2 correspond to the case of $k = 1/2$, and Theorem 2.4 can apply to them.

Furthermore, for g_1, by the definitions of $\beta_{k,0}$ and $\gamma_{k,0}$ together with (2.9) we know that $\beta_{k,0} = 0$, $\gamma_{k,0} = 2$, and then $\mu_{k,0}(t) = 2\sqrt{2t},\ t \in [0, T]$. Thus, it is enough that the μ in (2.16) is bigger than $2\sqrt{2T}$. And, for g_2, by the definitions of $\beta_{k,0}$ and $\gamma_{k,0}$ together with (2.9) we know that $\beta_{k,0} = 0$, $\gamma_{k,0} = 1$, and then $\mu_{k,0}(t) = \sqrt{2t},\ t \in [0, T]$. Thus, it is enough that the μ in (2.16) is bigger than $\sqrt{2T}$.

Example 2.7. Define for $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$,

$$g(\omega, t, y, z) := |B_t(\omega)| + y^2\mathbf{1}_{y\le 0} + |y|(\ln|y|)^{3/4}\mathbf{1}_{|y|>1} + 2|z|(\ln|z|)^{1/4}\mathbf{1}_{|z|>1}.$$

Then, g satisfies assumptions (EX1) with $c = 2$, (EX2) with $\alpha_\cdot = |B_\cdot|, \beta = 1, \gamma = 2, \delta = 3/4, \lambda = 1/4$ and (UN3). Consequently, this generator g corresponds to the case of $k = 3/4 \in (1/2, 1)$, and Theorem 2.4 can apply to it. Furthermore, by the definitions of $\beta_{k,0}$ and $\gamma_{k,0}$ together with (2.12) we know that $\beta_{k,0} = \beta = 1$, $\gamma_{k,0} = \gamma = 2$, and then $\mu_{k,0}(t) \le 4\sqrt{t/3} + t,\ t \in [0, T]$. Thus, it is enough that the μ in (2.16) is bigger than $4\sqrt{T/3} + T$.

Example 2.8. Define for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$,

$$g_1(\omega, t, y, z) := -2e^y + |y| \ln |y| \mathbf{1}_{|y|\le 1} + 3|z|\sqrt{|\ln |z||}\mathbf{1}_{|z|\le 1},$$

$$g_2(\omega, t, y, z) := |B_t(\omega)| + 2e^{-y} + |y| \ln |y| \mathbf{1}_{|y|>1} + 3|z|\sqrt{\ln |z|}\mathbf{1}_{|z|>1}$$

and

$$\begin{aligned} g(\omega, t, y, z) &:= g_1(\omega, t, y, z) + g_2(\omega, t, y, z) \\ &= |B_t(\omega)| + 2\left(e^{-y} - e^y\right) + |y| \ln |y| + 3|z|\sqrt{|\ln |z||}. \end{aligned}$$

Then, the generator g satisfies assumptions (EX1) with $c = 3$ and (EX2) with $\alpha_\cdot = |B_\cdot| + 1/e + 3/\sqrt{e} + 4$, $\beta = 1$, $\gamma = 3$, $\delta = 1$, $\lambda = 1/2$, the generator g_1 satisfies assumptions (UN1) with $\rho(x) = \bar{h}(x)$ and (UN2) with $\kappa(x) := 3h(x)$, and the generator g_2 satisfies (UN3), where the function $h(\cdot)$ is defined in Example 2.6 and the function $\bar{h}(\cdot)$ is defined as follows:

$$\bar{h}(x) := x|\ln x|\mathbf{1}_{0\le x\le \varepsilon} + \bar{h}'_-(\varepsilon)(x - \bar{h}(\varepsilon)), \quad x \in \mathbb{R}_+$$

with ε being a positive constant small enough. Thus, the generator g corresponds to the case of $k = 1$, and Theorem 2.4 can apply to it. Furthermore, by the definitions of $\beta_{k,0}$ and $\gamma_{k,0}$ together with (2.15) we know that $\beta_{k,0} = \beta = 1$, $\gamma_{k,0} = \gamma = 3$, and then $\mu_{k,0}(t) \le (3\sqrt{2t} + 11t/2)e^t$, $t \in [0, T]$. Thus, it is enough that the μ in (2.16) is bigger than $(3\sqrt{2T} + 11T/2)e^T$.

Example 2.9. Define for $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$,

$$g(\omega, t, y, z) := |B_t(\omega)|^2 - e^{2y} - 2|y| \ln |y|\mathbf{1}_{|y|>1} - 3|z|(\ln |z|)^2\mathbf{1}_{|z|>1}.$$

It is easy to verify that g satisfies assumptions (EX1) with $c = 3$, (EX2) with $\alpha_\cdot = |B_\cdot|^2+1$, $\beta = 2$, $\gamma = 3$, $\delta = 1$, $\lambda = 2$ and (UN3). Consequently, this generator g corresponds to the case of $k = 2\lambda = 4 \in (1, +\infty)$, and Theorem 2.4 can apply to it. Furthermore, by the definitions of $\beta_{k,0}$, $\gamma_{k,0}$ and $\bar{k}_\lambda$ together with (2.10) we know that $\beta_{k,0} = \beta = 2$, $\gamma_{k,0} = \gamma = 3$, $\bar{k}_\lambda = 32$, and then $\mu_{k,0}(t) = 18(e^{8t} - 1)$, $t \in [0, T]$. Thus, it is enough that the μ in (2.16) is bigger than $18(e^{8T} - 1)$.

Remark 2.10. Let $\bar{g} := g_1 + g$, where g_1 and g are defined in Examples 2.6 and 2.7, respectively. Then, $\bar{g}$ satisfies (EX2) with $(\delta, \lambda) := (1, 1/4)$ (resp. $(\delta, \lambda) := (3/4, 1/2)$). Consequently, Theorem 2.4 with $k = 1$ can apply to this generator g.

Furthermore, in the case of $\delta = 1$ and $\lambda = 1/4$, by the definitions of $\beta_{k,0}$ and $\gamma_{k,0}$ together with (2.15) we know that $\beta_{k,0} = \beta = 2$, $\gamma_{k,0} = 0$, and then $\mu_{k,0}(t) \le 2te^{2t}$, $t \in [0, T]$. Thus, it is enough that the μ in (2.16) is bigger than $2Te^{2T}$. And, in the case of $\delta = 3/4$ and $\lambda = 1/2$, by the definitions of $\beta_{k,0}$ and $\gamma_{k,0}$ together with (2.13) we know that $\beta_{k,0} = 0$, $\gamma_{k,0} = \gamma = 4$, and then $\mu_{k,0}(t) \le 4\sqrt{t} + 8t$, $t \in [0, T]$. Thus, it is enough that the μ in (2.16) is bigger than $4\sqrt{T} + 8T$.

2.4. Comparison theorems

In this subsection, we present three comparison theorems for BSDEs, which yields the uniqueness assertion in Theorem 2.4 in a straightforward way. We always assume that ξ and ξ' are two terminal conditions satisfying that $\mathbb{P}$-*a.s.*, $\xi \le \xi'$, g and g' are two generators which are jointly continuous in (y, z), and $(Y_t, Z_t)_{t\in[0,T]}$ and $(Y'_t, Z'_t)_{t\in[0,T]}$ are solutions to BSDE(ξ, g) and BSDE(ξ', g'), respectively.

Let the constant k be defined in (2.1), and the functions $\psi(\cdot, \cdot; k)$, $\mu_{k,\varepsilon}(\cdot)$ and $\mu_{k,0}(\cdot)$ be respectively defined in (2.2), (2.6) and (2.7) for $\varepsilon > 0$.

Theorem 2.11. *Assume that g (resp. g') satisfies assumptions (UN1) and (UN2), and $\mathbb{P}$-a.s.,*

$$\mathbf{1}_{Y_t>Y'_t}\left(g(t, Y'_t, Z'_t) - g'(t, Y'_t, Z'_t)\right) \le 0 \quad (\text{resp. } \mathbf{1}_{Y_t>Y'_t}\left(g(t, Y_t, Z_t) - g'(t, Y_t, Z_t)\right) \le 0 \text{ }). \tag{2.18}$$

If both processes $\psi(|Y_\cdot|, \mu_\cdot; k)$ and $\psi(|Y'_\cdot|, \mu_\cdot; k)$ belong to class (D) for some $k \ge 1/2$, where $\mu_\cdot$ be any nonnegative, strictly increasing and continuous function defined on $[0, T]$ with $\mu_0 = 0$, then $\mathbb{P}$-a.s., for each $t \in [0, T]$, $Y_t \le Y'_t$.

Theorem 2.12. *Assume that g (resp. g') satisfies assumptions (EX2) and (UN3), and $\mathbb{P}$-a.s.,*

$$g(t, Y'_t, Z'_t) \le g'(t, Y'_t, Z'_t) \quad (\text{resp. } g(t, Y_t, Z_t) \le g'(t, Y_t, Z_t) \text{ }). \tag{2.19}$$

If one of the following three assertions holds, then $\mathbb{P}$-a.s., $Y_t \le Y'_t$ for all $t \in [0, T]$:

(i) $k = \delta = 1/2$, and both $\psi(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, \mu_{k,0}(\cdot); k)$ and $\psi(|Y'_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, \mu_{k,0}(\cdot); k)$ belong to class (D);

(ii) $k \in [1/2, 1]$, and for some $\varepsilon > 0$, both $\psi(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, \mu_{k,\varepsilon}(\cdot); k)$ and $\psi(|Y'_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, \mu_{k,\varepsilon}(\cdot); k)$ belong to class (D);

(iii) $k > 1$, and for some $\varepsilon > 0$ and $q \ge 1$, both $\psi(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, 2q3^{k-1}\mu_{k,\varepsilon}(\cdot); k)$ and $\psi(|Y'_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, 2q3^{k-1}\mu_{k,\varepsilon}(\cdot); k)$ belongs to class (D).

Theorem 2.13. *Assume that the generator $g = g_1 + g_2$ (resp. $g' = g'_1 + g'_2$), g_1 (resp. g'_1) satisfies assumptions (UN1) and (UN2), g_2 (resp. g'_2) satisfies assumptions (EX2) and (UN3), and (2.19) holds. If $k = 1$ and there exists an $X \in L^1$ such that $\mathbb{P}$-a.s., for some $\varepsilon > 0$,*

$$\psi\left(|Y_t| + |Y'_t|, \mu_{k,\varepsilon}(t); k\right) \le \mathbb{E}[X|\mathcal{F}_t], \quad t \in [0, T], \tag{2.20}$$

then $\mathbb{P}$-a.s., $Y_t \le Y'_t$ for all $t \in [0, T]$.

Remark 2.14. In general, more assumptions on the generator g are required for the comparison theorem and uniqueness of the unbounded solutions to BSDEs than for the existence. Theorems 2.11–2.13 improve some related results of [5,11–13], and see Remark 2.3 for details. These three comparison theorems show the relevance of the comparison property of solutions of BSDEs, in addition to the usual (UN1)–(UN3), to the growth speed of the generators in the state variables y and z as well as the integrability condition of the solutions. Furthermore, it follows from Remark 2.2 that the generator g is of linear growth in the state variable z under (UN2). However, it is obvious that g can have a general growth in z under (UN3). This leads to that the integrability condition of the solutions required in Theorem 2.11 is weaker than that in Theorem 2.12, and (2.18) is also weaker than (2.19), see Theorem 2.1 and Remark 2.2 in [22] for more details.

2.5. An application in mathematical finance

For each $t \in [0, T]$, define

$$E^{\lambda}(\mathcal{F}_t) := \left\{ \xi \in \mathcal{F}_t \left| \mathbb{E}\left[\psi\left(|\xi|, \mu; (\lambda + \frac{1}{2}) \vee (2\lambda) \right) \right] < +\infty, \quad \forall \mu > 0 \right. \right\},$$

where the function $\psi(\cdot, \cdot; \cdot)$ is defined in (2.2). It is not difficult to verify that $E^{\lambda}(\mathcal{F}_t)$ is a linear space containing $L^{\infty}(\mathcal{F}_t)$, which is the set of all $\mathcal{F}_t$-measurable bounded random variables.

The following proposition is a direct consequence of Theorem 2.4, and the proof is omitted.

Proposition 2.15. *Assume that the generator $g(z) : \mathbb{R}^d \to \mathbb{R}$ is a concave function satisfying $g(0) = 0$ and*

$$|g(z)| \le a + \gamma |z| |\ln |z||^{\lambda} \tag{2.21}$$

with $a > 0$ being a given positive constant. Then, for each $\xi \in E^{\lambda}(\mathcal{F}_T)$, BSDE$(\xi, g)$ admits a unique solution $(Y_t, Z_t)_{t \in [0,T]}$ such that $Y_t \in E^{\lambda}(\mathcal{F}_t)$ for each $t \in [0, T]$.

Now, by virtue of Proposition 2.15, for each $\xi \in E^{\lambda}(\mathcal{F}_T)$ we can define

$$U_t^g(\xi) := Y_t, \quad t \in [0, T], \tag{2.22}$$

where $(Y_t, Z_t)_{t \in [0,T]}$ is the unique solution of BSDE(ξ, g) such that $Y_t \in E^{\lambda}(\mathcal{F}_t)$ for each $t \in [0, T]$.

The following theorem indicates that the family of operators $\{U_t^g(\cdot)\}_{t \in [0,T]}$ defined via (2.22) constitutes a dynamic utility process defined on $E^{\lambda}(\mathcal{F}_T)$.

Theorem 2.16. *For each $t \in [0, T]$, the mapping $U_t^g(\cdot) : E^{\lambda}(\mathcal{F}_T) \to E^{\lambda}(\mathcal{F}_t)$ defined via (2.22) satisfies the following properties:*

(i) **Positivity**: *$U_t^g(0) = 0$ and $U_t^g(\xi) \ge 0$ for each nonnegative random variable $\xi \in E^{\lambda}(\mathcal{F}_T)$;*

(ii) **Monotonicity**: *for each $\xi, \eta \in E^{\lambda}(\mathcal{F}_T)$, if $\xi \ge \eta$, then $U_t^g(\xi) \ge U_t^g(\eta)$;*

(iii) **Monetary**: *$U_t^g(\xi + \eta) = U_t^g(\xi) + \eta$ for each $\xi \in E^{\lambda}(\mathcal{F}_T)$ and $\eta \in E^{\lambda}(\mathcal{F}_t)$;*

(iv) **Concavity**: *$U_t^g(\theta\xi + (1-\theta)\eta) \ge \theta U_t^g(\xi) + (1-\theta) U_t^g(\eta)$ for all $\xi, \eta \in E^{\lambda}(\mathcal{F}_T)$ and $\theta \in (0, 1)$.*

Proof. In view of $g(0) = 0$ and the fact that g is independent of y, (i)–(iii) are the direct consequences of Proposition 2.15 and Theorem 2.12. Next we prove (iv). Let $(Y_t, Z_t)_{t \in [0,T]}$ and $(\bar{Y}_t, \bar{Z}_t)_{t \in [0,T]}$ be, respectively, the unique solution of BSDE(ξ, g) and BSDE(η, g) such that $Y_t, \bar{Y}_t \in E^{\lambda}(\mathcal{F}_t)$ for each $t \in [0, T]$. Then, $U_t^g(\xi) = Y_t$ and $U_t^g(\eta) = \bar{Y}_t$ for each $t \in [0, T]$, and it is clear that

$$(Y'_t, Z'_t)_{t \in [0,T]} := (\theta Y_t + (1-\theta)\bar{Y}_t, \theta Z_t + (1-\theta)\bar{Z}_t)_{t \in [0,T]}$$

solves the following BSDE

$$Y'_t = \theta\xi + (1-\theta)\eta + \int_t^T g'(s) \mathrm{d}s - \int_t^T Z'_s \cdot \mathrm{d}B_s, \quad t \in [0, T], \tag{2.23}$$

where $g'(t) := \theta g(Z_t) + (1-\theta) g(\bar{Z}_t),\ t \in [0, T]$. Furthermore, let $(\hat{Y}_t, \hat{Z}_t)_{t \in [0,T]}$ be the unique solution of BSDE$(\theta\xi + (1-\theta)\eta, g)$ such that $\hat{Y}_t \in E^{\lambda}(\mathcal{F}_t)$ for each $t \in [0, T]$, that is

$$\hat{Y}_t = \theta\xi + (1-\theta)\eta + \int_t^T g(\hat{Z}_s) \mathrm{d}s - \int_t^T \hat{Z}_s \cdot \mathrm{d}B_s, \quad t \in [0, T]. \tag{2.24}$$

Then, $U_t^g(\theta\xi+(1-\theta)\eta)=\hat{Y}_t$ for each $t\in[0,T]$. Since g a concave function satisfying (2.21), we have

$$g(Z_t')=g(\theta Z_t+(1-\theta)\bar{Z}_t)\geq\theta g(Z_t)+(1-\theta)g(\bar{Z}_t)=g'(t),\quad t\in[0,T].$$

It then follows from Theorem 2.12 that $\mathbb{P}$-$a.s.$, $\hat{Y}_t\geq Y_t'$ for all $t\in[0,T]$, which is the desired assertion. □

Now, let the function $f:\mathbb{R}^d\to\mathbb{R}_+$ be convex, satisfy $f(0)=0$, and grow in a super-quadratic way, i.e., $\liminf_{|x|\to\infty}f(x)/|x|^2>0$. For each $\xi\in L^\infty(\mathcal{F}_T)$, we define

$$\bar{U}_t(\xi):=\operatorname{essinf}\left\{\mathbb{E}_{\mathbb{Q}}\left[\xi+\int_t^T f(q_u)\mathrm{d}u\middle|\mathcal{F}_t\right]\middle|\mathbb{Q}\sim\mathbb{P}\right\},\quad t\in[0,T],\tag{2.25}$$

where $\mathbb{E}_{\mathbb{Q}}[\cdot|\mathcal{F}_t]$ is the expectation operator conditioned on $\mathcal{F}_t$ with respect to the probability measure $\mathbb{Q}$, which is equivalent to $\mathbb{P}$ and

$$\mathbb{E}\left[\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\middle|\mathcal{F}_t\right]=\exp\left\{\int_0^t q_u\mathrm{d}B_u-\frac{1}{2}\int_0^t|q_u|^2\mathrm{d}u\right\}.$$

It is not very hard to verify that $\{\bar{U}_t(\cdot)\}_{t\in[0,T]}$ defined by (2.25) constitutes a dynamic utility process defined on $L^\infty(\mathcal{F}_T)$. And, it follows from Theorems 2.1-2.2 in [48] that there exists a $(Z_t)_{t\in[0,T]}\in\mathcal{M}^2$ such that $(\bar{U}_t(\xi),Z_t)_{t\in[0,T]}$ is the unique bounded solution of the following BSDE

$$Y_t=\xi+\int_t^T\bar{f}(Z_s)\mathrm{d}s-\int_t^T Z_s\cdot\mathrm{d}B_s,\quad t\in[0,T],\tag{2.26}$$

where

$$\bar{f}(z):=-\sup_{x\in\mathbb{R}^d}(-z\cdot x-f(x))=\inf_{x\in\mathbb{R}^d}(z\cdot x+f(x))\leq 0,\quad\forall z\in\mathbb{R}^d\tag{2.27}$$

is a concave function with $\bar{f}(0)=0$, and has a sub-quadratic growth, i.e., $\limsup_{|x|\to\infty}|f(x)|/|x|^2<\infty$.

If $\gamma,\lambda>0$ and the function f satisfies

$$f(x)\geq\int_0^{|x|}\left\{\exp\left[\left(\frac{t}{\gamma}\right)^{\frac{1}{\lambda}}\right]-1\right\}\mathrm{d}t,\quad\forall x\in\mathbb{R}^d,\tag{2.28}$$

we deduce from (2.27) that for each $\varepsilon>0$, there exists a $c>0$ depending only on $(\gamma,\lambda,\varepsilon)$ such that

$$0\geq\bar{f}(z)\geq-\gamma|z|\,|\ln(1+|z|)|^\lambda\geq-c-(\gamma+\varepsilon\mathbf{1}_{\lambda>1})|z|\|\ln|z|\|^\lambda.\tag{2.29}$$

For example, by picking $\gamma=\lambda=1$ and

$$f(x)=\int_0^{|x|}(e^t-1)\mathrm{d}t=e^{|x|}-|x|-1\geq 0,\quad x\in\mathbb{R}^d,$$

we have

$$0\geq\bar{f}(z)=-|z|\ln(1+|z|)+|z|-\ln(1+|z|)\geq-|z|\ln(1+|z|)\geq-1-|z|\|\ln|z\|,\quad z\in\mathbb{R}^d.$$

In conclusion, if $f:\mathbb{R}^d\to\mathbb{R}_+$ is convex such that $f(0)=0$ and the inequality (2.28) is satisfied, then the function $\bar{f}$ defined by (2.27) is concave satisfying $\bar{f}(0)=0$ and (2.29). Moreover, for each $\xi\in L^\infty(\mathcal{F}_T)$, the dynamic utility process $\bar{U}_t(\xi)$ defined by (2.25) admits a dual representation $U_t^{\bar{f}}(\xi)$, which is defined by (2.22) via BSDE$(\xi,\bar{f})$.

Remark 2.17. It is well known that (2.25) is used to define the utility of bounded endowments in mathematical finance, see for example [48] for details. However, it is only defined on the space $L^\infty(\mathcal{F}_T)$. Our first aim is to extend the definition to a larger space to incorporate unbounded endowments. This motivates the definition of the dynamic utility process by (2.22) via a BSDE, which can be defined on a larger space $E^\lambda(\mathcal{F}_T)$ than $L^\infty(\mathcal{F}_T)$. To establish further the duality result in a larger space than $L^\infty(\mathcal{F}_T)$ will be the subject of our future research. Some relevant results on the link of the dynamic utility process with a BSDE are available in [5,58].

Remark 2.18. In a symmetric way to the above, it can be proved that if the generator g is a convex function and satisfies $g(0) = 0$ and (2.21), then for each $t \in [0, T]$, the mapping $U_t^g(\cdot) : E^\lambda(\mathcal{F}_T) \to E^\lambda(\mathcal{F}_t)$ defined by (2.22) via BSDE satisfies properties (i)–(iii) of Theorem 2.16 and convexity (instead of concavity)as well, and then specifies a convex risk measure on the space $E^\lambda(\mathcal{F}_T)$. Convex risk measures have been extensively investigated in the literature, see [59,60] among others.

3. Proof of main results

3.1. Proof of well-posedness of BSDEs (Theorem 2.4)

To prove Theorem 2.4, our whole idea is to establish some uniform a priori estimate and then apply the localization procedure put forward initially in [36].

We first introduce a general stability result for the solutions of BSDEs. For each pair of positive integers $n, p \geq 1$, set

$$\xi^{n,p} := \xi^+ \wedge n - \xi^- \wedge p \quad \text{and} \quad g^{n,p}(\omega, t, y, z) := g^+(\omega, t, y, z) \wedge n - g^-(\omega, t, y, z) \wedge p, \tag{3.1}$$

where and hereafter, a^+ stands for the maximum of a and 0, $a^- := (-a)^+$ and $a \wedge b$ represents the minimum of a and b. It follows from [35] that for each $n, p \geq 1$, the following BSDE$(\xi^{n,p}, g^{n,p})$ admits a minimal (maximal) bounded solution $(Y_t^{n,p}, Z_t^{n,p})_{t\in[0,T]}$ such that $Y_\cdot^{n,p}$ is bounded and $Z_\cdot^{n,p} \in \mathcal{M}^2$:

$$Y_t^{n,p} = \xi^{n,p} + \int_t^T g^{n,p}(s, Y_s^{n,p}, Z_s^{n,p})\mathrm{d}s - \int_t^T Z_s^{n,p} \cdot \mathrm{d}B_s, \quad t \in [0, T]. \tag{3.2}$$

By the comparison theorem, $Y_\cdot^{n,p}$ is nondecreasing in n and non-increasing in p. Furthermore, we have the following monotone stability result, which slightly generalizes those of [35,37]. Its proof is postponed in the Appendix.

Proposition 3.1. *Assume that ξ is a terminal condition and the generator g is continuous in the state variables (y, z) and satisfies (EX1). For each pair of positive integers $n, p \geq 1$, let $\xi^{n,p}$ and $g^{n,p}$ be defined in (3.1), and $(Y_\cdot^{n,p}, Z_\cdot^{n,p})$ be the minimal (maximal) bounded solution of (3.2). If there exists an $\mathbb{R}_+$-valued, progressively measurable and continuous process $(X_t)_{t\in[0,T]}$ such that*

$$\mathrm{d}\mathbb{P} \times \mathrm{d}t\text{-}a.e., \quad \forall n, p \geq 1, \quad |Y_\cdot^{n,p}| \leq X_\cdot,$$

then there exists an $\mathbb{R}^d$-valued adapted process $(Z_t)_{t\in[0,T]}$ such that $(Y_\cdot := \inf_p \sup_n Y_\cdot^{n,p},\ Z_\cdot)$ is a solution to BSDE(ξ, g).

Now, to apply the preceding stability theorem, we need the uniform a priori bound $X_\cdot$ for the processes $Y_\cdot^{n,p}$, which is crucial. Our strategy is to first search for an appropriate function $\phi(s, x)$ and then apply Itô-Tanaka's formula to $\phi(s, |Y_s^{n,p}|)$ on the time interval $s \in [t, \tau_m^t]$ with $t \in (0, T]$ and τ_m^t being an $(\mathcal{F}_t)$-stopping time valued in $[t, T]$. More specifically, we need to find a nonnegative continuous function $\phi(\cdot, \cdot) : [0, T] \times \mathbb{R}_+ \to \mathbb{R}_+$ such that $\phi_x(s, x) > 0$, $\phi_{xx}(s, x) > 0$ and

$$- \phi_x(s, x) \left(\beta x(\ln x)^\delta \mathbf{1}_{x>1} + \gamma |z| \|\ln |z|\|^\lambda\right) + \frac{1}{2}\phi_{xx}(s, x)|z|^2 + \phi_s(s, x) \geq 0 \tag{3.3}$$

for all $(s, x, z) \in (0, T] \times \mathbb{R}_+ \times \mathbb{R}^d$. Here and hereafter, ϕ_s is the first-order partial derivative of ϕ with respect to the first variable, and ϕ_x and ϕ_{xx} are the first- and second-order partial derivatives of ϕ with respect to the second variable.

To find the function ϕ satisfying inequality (3.3), our next idea is to find, for the term in the left hand side of (3.3), a perfect lower bound $\Phi(s, x)$ without containing the variable z, and transfer (3.3) to the inequality $\Phi(s, x) \geq 0$. For the case $\lambda = 0$, we use the inequality $2ab \leq a^2 + b^2$ to get that for each $(s, x, z) \in (0, T] \times \mathbb{R}_+ \times \mathbb{R}^d$,

$$\begin{aligned} -\gamma \phi_x(s, x)|z| + \frac{1}{2}\phi_{xx}(s, x)|z|^2 &= \frac{1}{2}\phi_{xx}(s, x)\left(-2\frac{\gamma\phi_x(s, x)}{\phi_{xx}(s, x)}|z| + |z|^2\right) \\ &\geq -\frac{\gamma^2}{2}\frac{(\phi_x(s, x))^2}{\phi_{xx}(s, x)}. \end{aligned}$$

Hence, (3.3) holds if the function $\phi(\cdot, \cdot)$ satisfies the following condition:

$$\forall (s, x) \in (0, T] \times \mathbb{R}_+, \quad -\beta\phi_x(s, x)x(\ln x)^\delta \mathbf{1}_{x>1} - \frac{\gamma^2}{2}\frac{(\phi_x(s, x))^2}{\phi_{xx}(s, x)} + \phi_s(s, x) \geq 0. \tag{3.4}$$

However, for the case of $\lambda > 0$, we need the following proposition, whose proof will be given in the Appendix.

Proposition 3.2. *Given $\lambda > 0$. For each $p > 1$, there is a positive constant $C_{p,\lambda} > 0$ depending only on (p, λ) such that*

$$\forall x, y > 0, \quad 2xy|\ln y|^\lambda \leq x^2\left(p4^{(\lambda-1)^+}|\ln x|^{2\lambda} + C_{p,\lambda}\right) + y^2. \tag{3.5}$$

Moreover, for each $p \leq 1/4^{(\lambda-1)^+}$, there is no constant $C_{p,\lambda}$ to satisfy the last inequality.

Remark 3.3. It is still open that what happens in Proposition 3.2 when $p \in (1/4^{(\lambda-1)^+}, 1]$.

With Proposition 3.2 in the hand, coming back to (3.3), observe that for each $\varepsilon > 0$, $\lambda > 0$ and $\gamma > 0$, there exists a constant $C_{\varepsilon,\lambda} > 0$ such that for each $(s, x, z) \in (0, T] \times \mathbb{R}_+ \times \mathbb{R}^d$,

$$\begin{aligned} &-\gamma\phi_x(s, x)|z|\|\ln |z|\|^\lambda + \frac{1}{2}\phi_{xx}(s, x)|z|^2 \\ =\ &\frac{1}{2}\phi_{xx}(s, x)\left(-2\frac{\gamma\phi_x(s, x)}{\phi_{xx}(s, x)}|z|\|\ln |z|\|^\lambda + |z|^2\right) \\ \geq\ &-\frac{\gamma^2}{2}\frac{(\phi_x(s, x))^2}{\phi_{xx}(s, x)}\left((1+\varepsilon)4^{(\lambda-1)^+}\left|\ln\frac{\gamma\phi_x(s, x)}{\phi_{xx}(s, x)}\right|^{2\lambda} + C_{\varepsilon,\lambda}\right). \end{aligned}$$

Then, (3.3) holds if for some $\varepsilon > 0$ and $C_{\varepsilon,\lambda} > 0$ associated with $p := 1+\varepsilon$ by Proposition 3.2, the function $\phi(s, x)$ satisfies that for each $(s, x) \in (0, T] \times \mathbb{R}_+$, with $\gamma > 0$,

$$
\begin{aligned}
&-\beta\phi_x(s,x)x(\ln x)^\delta \mathbf{1}_{x>1} - \frac{\gamma^2}{2}\frac{(\phi_x(s,x))^2}{\phi_{xx}(s,x)}\left((1+\varepsilon)4^{(\lambda-1)^+}\left|\ln\frac{\gamma\phi_x(s,x)}{\phi_{xx}(s,x)}\right|^{2\lambda} + C_{\varepsilon,\lambda}\right)\\
&\quad + \phi_s(s,x) \geq 0.
\end{aligned}
\tag{3.6}
$$

The following proposition gives the function $\phi(\cdot,\cdot)$ satisfying (3.4) and (3.6), and then (3.3), whose proof is placed in the Appendix. This proposition will play an important role in the proof of Theorem 2.4.

Proposition 3.4. *Let the constant k be defined in (2.1), the function $\mu_{k,0}(\cdot)$ be defined in (2.7) and $k_0 = e$. And, for $\varepsilon > 0$ let the function $\mu_{k,\varepsilon}(\cdot)$ be defined in (2.6) and*

$$
k_\varepsilon := \exp\left[\left(1+\frac{1}{\varepsilon}\right)^{\Delta}\right] + \left(\frac{1}{2\varepsilon^2}\right)^{\frac{1}{2\varepsilon}} + \frac{\mu_{k,\varepsilon}(T)}{\gamma} + \left(\frac{k\mu_{k,\varepsilon}(T)}{\gamma}\right)^2
$$

with

$$
\Delta := \frac{\mathbf{1}_{0\leq\delta<1/2}}{1/2-\delta} + \frac{\mathbf{1}_{\delta\neq 1}}{1-\delta} + \frac{\mathbf{1}_{0\leq\lambda<1/2}}{1/2-\lambda} + \frac{\mathbf{1}_{\delta\neq\lambda+1/2}}{|\lambda+1/2-\delta|} + \frac{\mathbf{1}_{\lambda>1/2}}{2\lambda-1}.
$$

Then, for each $\varepsilon > 0$ (resp. $\varepsilon \geq 0$ in the case of $k = \delta = 1/2$) there exists a strictly increasing and continuously differential function $\nu_{k,\varepsilon}(s) : [0, T] \to \mathbb{R}_+$ with $\nu_{k,\varepsilon}(0) = 0$ such that the following function

$$
\varphi(s,x;k,\varepsilon) := (x+k_\varepsilon)\exp\left(\mu_{k,\varepsilon}(s)\left(\ln(x+k_\varepsilon)\right)^k + \nu_{k,\varepsilon}(s)\right), \quad (s,x)\in[0,T]\times\mathbb{R}_+ \tag{3.7}
$$

satisfies (3.4) in case of $\lambda = 0$ and (3.6) in case of $\lambda > 0$, and then (3.3) for $\lambda \geq 0$.

The relationship between $\varphi(\cdot,\cdot;k,\varepsilon)$ in (3.7) and $\psi(\cdot,\cdot;k)$ in (2.2) is illustrated in the following proposition, whose proof is also postponed in the Appendix.

Proposition 3.5. *Let the constant k be defined in (2.1), and the functions $\mu_{k,\varepsilon}(\cdot)$, $\mu_{k,0}(\cdot)$, $\psi(\cdot,\cdot;k)$ and $\varphi(\cdot,\cdot;k,\varepsilon)$ be respectively defined on (2.6), (2.7), (2.2) and (3.7). Then, for each $\varepsilon \geq 0$ there exists a universal constant $C > 0$ depending on $(k, \beta, \gamma, \lambda, \varepsilon, T)$ such that*

$$
\forall (s,x)\in[0,T]\times\mathbb{R}_+, \quad \psi(x,\mu_{k,\varepsilon}(s);k) \leq \varphi(s,x;k,\varepsilon) \leq C\psi(x,\mu_{k,\varepsilon}(s);k) + C. \tag{3.8}
$$

Using Propositions 3.4 and 3.5, we can prove the following Proposition 3.6 on a priori bound X for the processes $Y^{n,p}$ needed in Proposition 3.1.

Proposition 3.6. *Let the constant k be defined in (2.1), the functions $\mu_{k,\varepsilon}(\cdot)$, $\mu_{k,0}(\cdot)$ and $\psi(\cdot,\cdot;k)$ be respectively defined in (2.6), (2.2) and (2.7), and ξ be a terminal condition. If the generator g satisfies (EX2), $|\xi| + \int_0^T \alpha_t \mathrm{d}t$ is a bounded random variable, and $(Y_t, Z_t)_{t\in[0,T]}$ is a solution of BSDE(ξ, g) such that $Y_\cdot$ is bounded, then for each $\varepsilon > 0$ (resp. $\varepsilon \geq 0$ in the case of $k = \delta = 1/2$), $\mathbb{P}$-a.s.,*

$$
\begin{aligned}
|Y_t| &\leq \psi\left(|Y_t| + \int_0^t \alpha_s \mathrm{d}s,\ \mu_{k,\varepsilon}(t);\ k\right)\\
&\leq C\mathbb{E}\left[\psi\left(|\xi| + \int_0^T \alpha_s \mathrm{d}s,\ \mu_{k,\varepsilon}(T);\ k\right)\middle|\mathcal{F}_t\right] + C, \quad t\in[0,T],
\end{aligned}
\tag{3.9}
$$

where C is a positive constant depending only on $(k, \beta, \gamma, \lambda, \varepsilon, T)$.

Proof. Define

$$\bar{Y}_t := |Y_t| + \int_0^t \alpha_s \mathrm{d}s \quad \text{and} \quad \bar{Z}_t := \mathrm{sgn}(Y_t) Z_t, \quad t \in [0, T].$$

Using Itô-Tanaka's formula, we have

$$\bar{Y}_t = \bar{Y}_T + \int_t^T (\mathrm{sgn}(Y_s) g(s, Y_s, Z_s) - \alpha_s)\, \mathrm{d}s - \int_t^T \bar{Z}_s \cdot \mathrm{d}B_s - \int_t^T \mathrm{d}L_s, \quad t \in [0, T],$$

with $L_\cdot$ being the local time of Y at the origin. Now, we fix $\varepsilon > 0$ (resp. $\varepsilon \geq 0$ in the case of $k = \delta = 1/2$) and apply Itô-Tanaka's formula to the process $\varphi(s, \bar{Y}_s; k, \varepsilon)$ (see (3.7) for the definition of the function $\varphi(\cdot, \cdot; k, \varepsilon)$), to derive, in view of (EX2),

$$\begin{aligned}
\mathrm{d}\varphi(s, \bar{Y}_s; k, \varepsilon) &= \varphi_x(s, \bar{Y}_s; k, \varepsilon) \left(-\mathrm{sgn}(Y_s) g(s, Y_s, Z_s) + \alpha_s\right) \mathrm{d}s + \varphi_x(s, \bar{Y}_s; k, \varepsilon) \bar{Z}_s \cdot \mathrm{d}B_s \\
&\quad + \varphi_x(s, \bar{Y}_s; k, \varepsilon) \mathrm{d}L_s + \frac{1}{2} \varphi_{xx}(s, \bar{Y}_s; k, \varepsilon) |Z_s|^2 \mathrm{d}s + \varphi_s(s, \bar{Y}_s; k, \varepsilon) \mathrm{d}s \\
&\geq \Bigg[-\varphi_x(s, \bar{Y}_s; k, \varepsilon) \left(\beta |Y_s| (\ln |Y_s|)^\delta \mathbf{1}_{|Y_s|>1} + \gamma |Z_s| \|\ln |Z_s|\|^\lambda\right) \\
&\quad + \frac{1}{2} \varphi_{xx}(s, \bar{Y}_s; k, \varepsilon) |Z_s|^2 + \varphi_s(s, \bar{Y}_s; k, \varepsilon) \Bigg] \mathrm{d}s \\
&\quad + \varphi_x(s, \bar{Y}_s; k, \varepsilon) \bar{Z}_s \cdot \mathrm{d}B_s, \quad s \in [0, T].
\end{aligned}$$

Furthermore, in view of the fact that

$$|Y_s| (\ln |Y_s|)^\delta \mathbf{1}_{|Y_s|>1} \leq \bar{Y}_s (\ln \bar{Y}_s)^\delta \mathbf{1}_{\bar{Y}_s > 1}$$

and Proposition 3.4, we have

$$\mathrm{d}\varphi(s, \bar{Y}_s; k, \varepsilon) \geq \varphi_x(s, \bar{Y}_s; k, \varepsilon) \bar{Z}_s \cdot \mathrm{d}B_s, \quad s \in [0, T]. \tag{3.10}$$

Let us consider, for each integer $n \geq 1$ and each $t \in [0, T]$, the following stopping time

$$\tau_n^t := \inf \left\{ s \in [t, T] : \int_t^s \left[\varphi_x(r, \bar{Y}_r; k, \varepsilon)\right]^2 |\bar{Z}_r|^2 \mathrm{d}r \geq n \right\} \wedge T$$

with the convention that $\inf \emptyset = +\infty$. It follows from the definition of τ_n^t and the inequality (3.10) that

$$\forall n \geq 1, \quad \varphi(t, \bar{Y}_t; k, \varepsilon) \leq \mathbb{E}\left[\varphi(\tau_n^t, \bar{Y}_{\tau_n^t}; k, \varepsilon) \middle| \mathcal{F}_t \right], \quad t \in [0, T].$$

Thus, by Proposition 3.5, there exists a constant $C > 0$ depending only on $(k, \beta, \gamma, \lambda, \varepsilon, T)$ such that

$$\begin{aligned}
\psi(\bar{Y}_t, \ \mu_{k,\varepsilon}(t); \ k) &\leq \varphi(t, \bar{Y}_t; k, \varepsilon) \leq \mathbb{E}\left[\varphi(\tau_n^t, \bar{Y}_{\tau_n^t}; k, \varepsilon) \middle| \mathcal{F}_t \right] \\
&\leq C \mathbb{E}\left[\psi(\bar{Y}_{\tau_n^t}, \ \mu_{k,\varepsilon}(\tau_n^t); \ k) \middle| \mathcal{F}_t \right] + C, \quad t \in [0, T],
\end{aligned}$$

and then for each $n \geq 1$ and each $t \in [0, T]$, we have

$$\begin{aligned}
|Y_t| &\leq \psi \left(|Y_t| + \int_0^t \alpha_s \mathrm{d}s, \ \mu_{k,\varepsilon}(t); \ k \right) \\
&\leq C \mathbb{E}\left[\psi \left(|Y_{\tau_n^t}| + \int_0^{\tau_n^t} \alpha_s \mathrm{d}s, \ \mu_{k,\varepsilon}(\tau_n^t); \ k \right) \middle| \mathcal{F}_t \right] + C,
\end{aligned}$$

from which the desired inequality (3.9) follows by sending n to infinity. The proof is completed. □

Remark 3.7. From the preceding proof, assertions of Proposition 3.6 are still true when $(|\xi|, |Y_t|)$ is replaced with (ξ^+, Y_t^+), and (EX2) is replaced with the following one:

(EX2') It holds that $\mathrm{d}\mathbb{P} \times \mathrm{d}t$-$a.e.$,

$$\forall (y,z) \in \mathbb{R}_+ \times \mathbb{R}^d, \quad g(\omega,t,y,z) \le \alpha_t(\omega) + \beta|y|(\ln|y|)^\delta \mathbf{1}_{|y|>1} + \gamma|z|\|\ln|z|\|^\lambda.$$

In fact, it is sufficient to replace $(|Y_\cdot|, \mathrm{sgn}(Y_\cdot), L_\cdot)$ with $(Y_\cdot^+, \mathbf{1}_{Y_\cdot>0}, \frac{1}{2}L_\cdot)$ in the proof.

By virtue of Proposition 3.2 and Itô's formula, we can verify in the Appendix the following proposition, which will be used to prove Theorem 2.4.

Proposition 3.8. *Assume that the generator g satisfies (EX2) and $(Y_t, Z_t)_{t\in[0,T]}$ is a solution of BSDE(ξ, g). If $|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s \in \mathcal{S}^p$ for $p > 0$, then $Z_\cdot \in \mathcal{M}^q$ for each $q \in (0, p)$.*

Proof of Theorem 2.4. For each pair of positive integers $n, p \ge 1$, let $\xi^{n,p}$ and $g^{n,p}$ be defined in (3.1), and $(Y_\cdot^{n,p}, Z_\cdot^{n,p})$ be the minimal (maximal) bounded solution of (3.2). It is easy to check that the generator $g^{n,p}$ satisfies (EX2) with $\alpha_\cdot \wedge (n \vee p)$ instead of $\alpha_\cdot$.

Now, we assume that there exists a positive constant $\mu > \mu_{k,0}(T)$ (resp. $\mu \ge \mu_{k,0}(T)$ in the case of $k = \delta = 1/2$) such that (2.16) holds. From the definitions of $\mu_{k,\varepsilon}(\cdot)$ and $\mu_{k,0}(\cdot)$ in (2.6) and (2.7), it is not very difficult to find a (unique) positive constant $\varepsilon > 0$ (resp. $\varepsilon \ge 0$ in the case of $k = \delta = 1/2$) satisfying $\mu_{k,\varepsilon}(T) = \mu$. Then, applying Proposition 3.6 with this ε to BSDE (3.2) yields the existence of a constant $C > 0$ depending only on $(k, \beta, \gamma, \lambda, \varepsilon, T)$ such that $\mathbb{P}$-$a.s.$, for all $n, p \ge 1$,

$$\begin{aligned}
|Y_t^{n,p}| &\le \psi\left(|Y_t^{n,p}| + \int_0^t [\alpha_s \wedge (n \vee p)]\mathrm{d}s,\ \mu_{k,\varepsilon}(t);\ k\right) \\
&\le C\mathbb{E}\left[\psi\left(|\xi^{n,p}| + \int_0^T [\alpha_s \wedge (n \vee p)]\mathrm{d}s,\ \mu_{k,\varepsilon}(T);\ k\right)\Bigg|\mathcal{F}_t\right] + C \qquad (3.11)\\
&= C\mathbb{E}\left[\psi\left(|\xi| + \int_0^T \alpha_s \mathrm{d}s,\ \mu;\ k\right)\Bigg|\mathcal{F}_t\right] + C =: X_t, \qquad t \in [0,T].
\end{aligned}$$

Thus, in view of (2.16), we have found an $\mathbb{R}_+$-valued, progressively measurable and continuous process $(X_t)_{t\in[0,T]}$ such that

$$\mathrm{d}\mathbb{P} \times \mathrm{d}t\text{-}a.e., \quad \forall n, p \ge 1, \quad |Y_\cdot^{n,p}| \le X_\cdot.$$

Now, we can apply Proposition 3.1 to obtain the existence of an $\mathbb{R}^d$-valued adapted process $(Z_t)_{t\in[0,T]}$ such that $(Y_\cdot := \inf_p \sup_n Y_\cdot^{n,p},\ Z_\cdot)$ is a solution to BSDE(ξ, g). And, sending n, p to infinity in (3.11) yields the desired inequality (2.17), and then $\psi\left(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, \mu_{k,\varepsilon}(\cdot); k\right)$ belongs to class (D).

Furthermore, suppose that $k = 2\lambda > 1$ and (2.16) holds for some $\mu > 2q3^{k-1}\mu_{k,0}(T)$ with $q \ge 1$. Then, there exists a unique constant $\varepsilon > 0$ such that $2q3^{k-1}\mu_{k,\varepsilon}(T) = \mu$. In view of the definition of $\mu_{k,\varepsilon}(\cdot)$ and by virtue of the proof of Proposition 3.4, it is not very difficult to verify that Propositions 3.4 and 3.6 still hold when the function $\mu_{k,\varepsilon}(\cdot)$ is replaced with $2q3^{k-1}\mu_{k,\varepsilon}(\cdot)$, and then by the above analysis BSDE(ξ, g) admits a solution $(Y_\cdot, Z_\cdot)$ such that the process

$$\psi\left(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s,\ 2q3^{k-1}\mu_{k,\varepsilon}(\cdot);\ k\right)$$

belongs to class (D). And, (2.17) holds with $2q3^{k-1}\mu_{k,\varepsilon}(\cdot)$ instead of $\mu_{k,\varepsilon}(\cdot)$.

For the existence part of Theorem 2.4, it remains to show that the solution $(Y_\cdot, Z_\cdot)$ belongs to the desired space. It has the following three cases: $k \in [1/2, 1)$, $k = 1$ and $k > 1$. For the first case, in view of (2.3) and (2.17), we deduce from Lemma 6.1 in [16] that for some $\varepsilon > 0$ (resp. $\varepsilon \geq 0$ in the case of $k = \delta = 1/2$) and each $p \in (0, 1)$,

$$\begin{aligned}\mathbb{E}\left[\sup_{t\in[0,T]}\left(|Y_t| + \int_0^t \alpha_s \mathrm{d}s\right)^p\right] &\leq \mathbb{E}\left[\sup_{t\in[0,T]}\left(\psi\left(|Y_t| + \int_0^t \alpha_s \mathrm{d}s;\ \mu_{k,\varepsilon}(t);\ k\right)\right)^p\right]\\ &\leq \frac{1}{1-p}\left(\mathbb{E}\left[\psi\left(|\xi| + \int_0^T \alpha_s \mathrm{d}s;\ \mu_{k,\varepsilon}(T);\ k\right)\right]\right)^p\\ &< +\infty.\end{aligned}$$

For the case of $k = 1$, in view of (2.17), (2.4) and the fact that $\mu_{k,\varepsilon}(\cdot) \geq \varepsilon$, we deduce from Lemma 6.1 in [16] that for some $\varepsilon > 0$ and each $p \in (0, 1)$,

$$\begin{aligned}\mathbb{E}\left[\sup_{t\in[0,T]}\left(|Y_t| + \int_0^t \alpha_s \mathrm{d}s\right)^{p(1+\varepsilon)}\right] &\leq \mathbb{E}\left[\sup_{t\in[0,T]}\left(\psi\left(|Y_t| + \int_0^t \alpha_s \mathrm{d}s;\ \mu_{k,\varepsilon}(t);\ k\right)\right)^p\right]\\ &\leq \frac{1}{1-p}\left(\mathbb{E}\left[\psi\left(|\xi| + \int_0^T \alpha_s \mathrm{d}s;\ \mu_{k,\varepsilon}(T);\ k\right)\right]\right)^p\\ &< +\infty.\end{aligned}$$

For the last case of $k > 1$, in view of (2.17), (2.5) and the fact that $\mu_{k,\varepsilon}(\cdot) \geq \varepsilon$, we deduce from Lemma 6.1 in [16] that for some $\varepsilon > 0$ and each $p > 1$,

$$\begin{aligned}\mathbb{E}\left[\sup_{t\in[0,T]}\left(|Y_t| + \int_0^t \alpha_s \mathrm{d}s\right)^p\right] &\leq \mathbb{E}\left[\sup_{t\in[0,T]}\left(\psi\left(|Y_t| + \int_0^t \alpha_s \mathrm{d}s;\ \mu_{k,\varepsilon}(t);\ k\right)\right)^{1/p}\right]\\ &\leq \frac{p}{p-1}\left(\mathbb{E}\left[\psi\left(|\xi| + \int_0^T \alpha_s \mathrm{d}s;\ \mu_{k,\varepsilon}(T);\ k\right)\right]\right)^{1/p}\\ &< +\infty.\end{aligned}$$

Then, the conclusion that $(Y_\cdot, Z_\cdot)$ belongs to the desired space follows from Proposition 3.8 immediately. The existence part is proved.

Finally, the uniqueness is a direct consequence of Comparison Theorems 2.11 to 2.13, which will be proved in the next subsection. $\square$

3.2. *Proof of comparison (Theorems 2.12–2.13)*

To prove Theorem 2.11, we need the following two lemmas, which can be proved in the same way as in Hu and Tang [10].

Lemma 3.9. *Let the function $\psi(\cdot, \cdot; k)$ with some $k \in [1/2, 1]$ be defined in (2.2). Then we have*

(i) For each $x \in \mathbb{R}_+$, $\psi(x, \cdot; k)$ is increasing on $\mathbb{R}_+$;
(ii) For $\mu \in \mathbb{R}_+$, $\psi(\cdot, \mu; k)$ is a positive, increasing and strictly convex function on $\mathbb{R}_+$;
(iii) For all $c > 1$ and $x, \mu \in \mathbb{R}_+$, $\psi(cx, \mu; k) \leq \psi(c, \mu; k)\psi(x, \mu; k)$;
(iv) For all $x_1, x_2, \mu \in \mathbb{R}_+$, $\psi(x_1 + x_2, \mu; k) \leq \frac{1}{2}\psi(2, \mu; k)\left[\psi(x_1, \mu; k) + \psi(x_2, \mu; k)\right]$;
(v) For each $x \in \mathbb{R}$, $y \in \mathbb{R}_+$ and $\mu > 0$, $e^x y \leq e^{\frac{x^2}{2\mu^2}} + e^{2\mu^2}\psi(y, \mu; \frac{1}{2})$.

Lemma 3.10. *Let $(q_t)_{t\in[0,T]}$ be an $\mathbb{R}^d$-valued progressively measurable process with $|q_\cdot| \le \epsilon$ almost surely. For each $t \in [0,T]$, if $0 \le \lambda < \frac{1}{2\epsilon^2(T-t)}$, then*

$$\mathbb{E}\left[e^{\lambda \left| \int_t^T q_s \cdot \mathrm{d}B_s \right|^2} \middle| \mathcal{F}_t \right] \le \frac{1}{\sqrt{1-2\lambda\epsilon^2(T-t)}}.$$

Proof of Theorem 2.11. We only prove the case that g verifies assumptions (UN1) and (UN2), and $\mathrm{d}\mathbb{P}\times\mathrm{d}t$-$a.e.$, $\mathbf{1}_{Y_t>Y'_t}\left(g(t,Y'_t,Z'_t)-g'(t,Y'_t,Z'_t)\right) \le 0$. The other case can be proved in an identical way.

Define $\hat{Y}_\cdot := Y_\cdot - Y'_\cdot$ and $\hat{Z}_\cdot := Z_\cdot - Z'_\cdot$. Then the pair $(\hat{Y}_\cdot, \hat{Z}_\cdot)$ verifies

$$\hat{Y}_t = \xi - \xi' + \int_t^T \left(g(s,Y_s,Z_s) - g'(s,Y'_s,Z'_s)\right)\mathrm{d}s - \int_t^T \hat{Z}_s\cdot\mathrm{d}B_s, \qquad t\in[0,T]. \tag{3.12}$$

From the assumption of $\mathbf{1}_{Y_t>Y'_t}\left(g(t,Y'_t,Z'_t)-g'(t,Y'_t,Z'_t)\right) \le 0$ and assumptions (UN1) and (UN2) of the generator g together with Remark 2.2, it is not difficult to deduce that $\mathrm{d}\mathbb{P}\times\mathrm{d}s$-$a.e.$,

$$\mathbf{1}_{\hat{Y}_s>0}\left(g(s,Y_s,Z_s)-g'(s,Y'_s,Z'_s)\right) \le \rho(\hat{Y}_s^+) + \mathbf{1}_{\hat{Y}_s>0}\kappa(|\hat{Z}_s|) \le A\hat{Y}_s^+ + A\mathbf{1}_{\hat{Y}_s>0}|\hat{Z}_s| + 2A. \tag{3.13}$$

For each $t\in[0,T]$ and each integer $n\ge 1$, define the following stopping times:

$$\sigma_n^t := \inf\left\{ s\in[t,T]:\ |\hat{Y}_s| + \int_t^s |\hat{Z}_r|^2\mathrm{d}r \ge n \right\}\wedge T$$

with the convention that $\inf\emptyset = +\infty$. Itô-Tanaka's formula applied to (3.12) gives

$$\begin{aligned}\hat{Y}_t^+ = \hat{Y}^+_{\sigma_n^t} &+ \int_t^{\sigma_n^t} \mathbf{1}_{\hat{Y}_s>0}\left(g(s,Y_s,Z_s)-g'(s,Y'_s,Z'_s)\right)\mathrm{d}s \\ &- \int_t^{\sigma_n^t}\mathbf{1}_{\hat{Y}_s>0}\hat{Z}_s\cdot\mathrm{d}B_s - \int_t^{\sigma_n^t}\mathrm{d}L_s, \qquad t\in[0,T],\end{aligned}$$

where $L_\cdot$ is the local time of $\hat{Y}_\cdot$ at the origin. Then, in view of (3.13),

$$e^{At}\hat{Y}_t^+ \le e^{AT}(\hat{Y}^+_{\sigma_n^t} + 2AT) + \int_t^{\sigma_n^t} v_s\cdot\tilde{Z}_s\mathrm{d}s - \int_t^{\sigma_n^t}\tilde{Z}_s\cdot\mathrm{d}B_s, \qquad t\in[0,T], \tag{3.14}$$

with $v_s := A\frac{\hat{Z}_s}{|\hat{Z}_s|}\mathbf{1}_{|\hat{Z}_s|>0}$ and $\tilde{Z}_s := \mathbf{1}_{\hat{Y}_s>0}e^{As}\hat{Z}_s$. It is clear that $|v_\cdot|\le A$ and it follows from (3.14) that

$$\forall n\ge 1,\quad \hat{Y}_t^+ \le e^{AT}\left(\mathbb{E}\left[e^{\int_t^{\sigma_n^t} v_s\cdot\mathrm{d}B_s}\hat{Y}^+_{\sigma_n^t}\middle|\mathcal{F}_t\right] + 2AT\right), \qquad t\in[0,T]. \tag{3.15}$$

Furthermore, in view of assertion (v) of Lemma 3.9, we know that for each $n\ge 1$,

$$e^{\int_t^{\sigma_n^t} v_s\cdot\mathrm{d}B_s}\hat{Y}^+_{\sigma_n^t} \le e^{\frac{1}{2\mu_t^2}\left(\int_t^{\sigma_n^t} v_s\cdot\mathrm{d}B_s\right)^2} + e^{2\mu_t^2}\psi\left(\hat{Y}^+_{\sigma_n^t}, \mu_t; \frac{1}{2}\right), \qquad t\in(0,T]. \tag{3.16}$$

And, by virtue of the assumptions of $\mu_\cdot$, we can pick a (unique) $T_1\in(0,T)$ such that $\mu^2_{T_1} = 4A^2(T-T_1)$ and $\mu_t^2 \ge 4A^2(T-t)$ for each $t\in[T_1,T]$. It then follows from Lemma 3.10

that for all $n \geq 1$,

$$\mathbb{E}\left[\left|e^{\frac{1}{2\mu_t^2}\left(\int_t^{\sigma_n^t} v_s \cdot \mathrm{d}B_s\right)^2}\right|^2\right] = \mathbb{E}\left[e^{\frac{1}{\mu_t^2}\left(\int_t^{\sigma_n^t} v_s \cdot \mathrm{d}B_s\right)^2}\right] \leq \frac{1}{\sqrt{1 - \frac{2A^2(T-t)}{\mu_t^2}}} \leq \sqrt{2}, \quad t \in [T_1, T],$$

and then, the family of random variables

$$\exp\left(\frac{1}{2\mu_t^2}\left(\int_t^{\sigma_n^t} v_s \cdot \mathrm{d}B_s\right)^2\right)$$

is uniformly integrable on the interval $[T_1, T]$. On the other hand, in view of Lemma 3.9 and the monotonicity of $\mu_\cdot$, observe that for all $n \geq 1$,

$$\begin{aligned} e^{2\mu_t^2}\psi\left(\hat{Y}^+_{\sigma_n^t}, \mu_t;\ \frac{1}{2}\right) &\leq e^{2\mu_T^2}\psi\left(\left|Y_{\sigma_n^t}\right| + \left|Y'_{\sigma_n^t}\right|, \mu_{\sigma_n^t};\ \frac{1}{2}\right) \\ &\leq \frac{e^{2\mu_T^2}\psi(2, \mu_T; 1/2)}{2}\left[\psi\left(\left|Y_{\sigma_n^t}\right|, \mu_{\sigma_n^t};\ \frac{1}{2}\right) + \psi\left(\left|Y'_{\sigma_n^t}\right|, \mu_{\sigma_n^t};\ \frac{1}{2}\right)\right], \quad t \in [0, T]. \end{aligned}$$

It then follows from (3.16) together with the assumption of Theorem 2.11 that for $t \in [T_1, T]$, the family of random variables $e^{\int_t^{\sigma_n^t} v_s \cdot \mathrm{d}B_s}\hat{Y}^+_{\sigma_n^t}$ is uniformly integrable. Consequently, in view of the fact that $\hat{Y}^+_T = (\xi - \xi')^+ = 0$, by letting $n \to \infty$ in (3.15) we get that $\hat{Y}_\cdot \leq 2ATe^{AT}$ on the time interval $[T_1, T]$, which means that $\hat{Y}_\cdot$ is a bounded process on $[T_1, T]$. Thus, by virtue of the assumptions of Theorem 2.11 again, we can apply Theorem 2.1 in [22] to obtain $\hat{Y}^+_\cdot = 0$ on the time interval $[T_1, T]$.

Now, in view of the fact $Y^+_{T_1} = 0$ obtained above, we can respectively replace T and T_1 in the above analysis with T_1 and $T_2 \in (0, T_1)$ satisfying $\mu_{T_2}^2 = 4A^2(T_1 - T_2)$ and $\mu_t^2 \geq 4A^2(T_1 - t)$ for each $t \in [T_2, T_1]$, and use the same argument to get that $\hat{Y}^+_\cdot = 0$ on the time interval $[T_2, T_1]$.

Finally, repeating the above procedure, we successively obtain that $\hat{Y}^+_\cdot = 0$ on the time intervals $[T_3, T_2], \cdots, [T_{m+1}, T_m], \cdots$, where $0 < T_{m+1} < T_m < T$ for each $m \geq 1$, and the limit of sequence $(T_m)_{m\geq 1}$ must exist and is denoted by $\bar{T}$. Noticing that $\mu_{T_m}^2 = 4A^2(T_{m-1} - T_m)$ by the above construction and in view of the assumptions of $\mu_\cdot$, we can conclude that $\bar{T} = \lim_{m\to\infty} T_m = 0$. Thus, in view of the continuity of process $\hat{Y}^+_t$ with respect to the time variable t, we have $\hat{Y}^+_\cdot = 0$ on the whole time interval $[0, T]$ by sending $m \to \infty$, which is the desired result. The proof is then complete. □

Proof of Theorem 2.12. To take advantage of the convexity (concavity) condition of the generator g, we use the θ-technique developed for example in [37]. More precisely, instead of estimating the difference between the processes $Y_\cdot$ and $Y'_\cdot$, we estimate $Y_\cdot - \theta Y'_\cdot$ for each $\theta \in (0, 1)$.

(i) Case of $k = \delta = 1/2$. We first consider the case that the generator g is convex in the state variables (y, z), satisfies (EX2), and $\mathrm{d}\mathbb{P} \times \mathrm{d}t$-$a.e.$,

$$g(t, Y'_t, Z'_t) \leq g'(t, Y'_t, Z'_t).$$

For each fixed $\theta \in (0, 1)$, define

$$\Delta^\theta U_\cdot := \frac{Y_\cdot - \theta Y'_\cdot}{1-\theta} \quad \text{and} \quad \Delta^\theta V_\cdot := \frac{Z_\cdot - \theta Z'_\cdot}{1-\theta}. \tag{3.17}$$

Then the pair $(\Delta^\theta U_\cdot, \Delta^\theta V_\cdot)$ satisfies the following BSDE:

$$\Delta^\theta U_t = \Delta^\theta U_T + \int_t^T \Delta^\theta g(s, \Delta^\theta U_s, \Delta^\theta V_s)\mathrm{d}s - \int_t^T \Delta^\theta V_s \cdot \mathrm{d}B_s, \qquad t \in [0, T], \tag{3.18}$$

where $\mathrm{d}\mathbb{P} \times \mathrm{d}s$-$a.e.$, for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$,

$$\begin{aligned} \Delta^\theta g(s, y, z) \quad := \quad & \frac{1}{1-\theta}\left[\ g(s, (1-\theta)y + \theta Y'_s, (1-\theta)z + \theta Z'_s) - \theta g(s, Y'_s, Z'_s)\ \right] \\ & + \frac{\theta}{1-\theta}\left[\ g(s, Y'_s, Z'_s) - g'(s, Y'_s, Z'_s)\ \right]. \end{aligned} \tag{3.19}$$

It follows from the assumptions that $\mathrm{d}\mathbb{P} \times \mathrm{d}s$-$a.e.$,

$$\forall (y, z) \in \mathbb{R}_+ \times \mathbb{R}^d, \quad \Delta^\theta g(s, y, z) \le g(s, y, z) \le \alpha_s + \beta|y|\sqrt{\ln|y|}\mathbf{1}_{|y|>1} + \gamma|z|, \tag{3.20}$$

which means that the generator $\Delta^\theta g$ satisfies (EX2') defined in Remark 3.7.

On the other hand, since both processes $\psi\left(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s,\ \mu_{k,0}(\cdot); k\right)$ and $\psi\left(|Y'_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s,\ \mu_{k,0}(\cdot); k\right)$ belong to class (D), by virtue of Lemma 3.9, we conclude that the process

$$\psi\left(\left(\Delta^\theta U_t\right)^+ + \int_0^t \alpha_s \mathrm{d}s,\ \ \mu_{k,0}(t);\ \ k\right), \qquad t \in [0, T]$$

belongs to class (D) for each $\theta \in (0, 1)$. Thus, for BSDE (3.18), by virtue of (3.20), Remark 3.7 and the proof of Proposition 3.6, we derive that there exists a constant $C > 0$ depending on $(k, \beta, \gamma, \lambda, T)$ such that $\mathbb{P}$-$a.s.$, for each $t \in [0, T]$,

$$\begin{aligned} \left(\Delta^\theta U_t\right)^+ &\le \psi\left(\left(\Delta^\theta U_t\right)^+,\ \mu_{k,0}(t);\ \ k\right) \\ &\le C\mathbb{E}\left[\psi\left(\left(\Delta^\theta U_T\right)^+ + \int_0^T \alpha_s \mathrm{d}s,\ \mu_{k,0}(T);\ \ k\right)\middle|\mathcal{F}_t\right] + C. \end{aligned} \tag{3.21}$$

Furthermore, since

$$\left(\Delta^\theta U_T\right)^+ = \frac{(\xi - \theta\xi')^+}{1-\theta} = \frac{\left[\xi - \theta\xi + \theta(\xi - \xi')\right]^+}{1-\theta} \le \xi^+, \tag{3.22}$$

we have from (3.21) that

$$\left(Y_t - \theta Y'_t\right)^+ \le (1-\theta)\left(C\mathbb{E}\left[\psi\left(\xi^+ + \int_0^T \alpha_s \mathrm{d}s,\ \ \mu_{k,0}(T);\ \ k\right)\middle|\mathcal{F}_t\right] + C\right), \qquad t \in [0, T].$$

Thus, the desired conclusion follows in the limit as $\theta \to 1$.

For the case that the generator g is concave in the state variables (y, z), we need to respectively use the $\theta Y_\cdot - Y'_\cdot$ and $\theta Z_\cdot - Z'_\cdot$ to replace $Y_\cdot - \theta Y'_\cdot$ and $Z_\cdot - \theta Z'_\cdot$ in (3.17). In this case, the generator $\Delta^\theta g$ in (3.19) should be replaced with

$$\begin{aligned} \Delta^\theta g(s, y, z) := & \frac{1}{1-\theta}\left[\ \theta g(s, Y_s, Z_s) - g(s, -(1-\theta)y + \theta Y_s, -(1-\theta)z + \theta Z_s)\ \right] \\ & + \frac{1}{1-\theta}\left[\ g(s, Y'_s, Z'_s) - g'(s, Y'_s, Z'_s)\ \right]. \end{aligned}$$

Since g is concave in (y, z), we have $\mathrm{d}\mathbb{P} \times \mathrm{d}s$-$a.e.$,

$$\forall (y, z) \in \mathbb{R} \times \mathbb{R}^d, \quad g(s, -(1-\theta)y + \theta Y_s, -(1-\theta)z + \theta Z_s) \ge \theta g(s, Y_s, Z_s) + (1-\theta)g(t, -y, -z),$$

and then, (3.20) can be replaced by

$$\forall (y,z) \in \mathbb{R}_+ \times \mathbb{R}^d, \quad \Delta^\theta g(s,y,z) \le -g(s,-y,-z) \le \alpha_s + \beta |y| \sqrt{\ln |y|} \mathbf{1}_{|y|>1} + \gamma |z|,$$

which means that the generator $\Delta^\theta g$ still satisfies (EX2'). Consequently, (3.21) still holds. Moreover, we use

$$\left(\Delta^\theta U_T\right)^+ = \frac{(\theta\xi - \xi')^+}{1-\theta} = \frac{\left[\theta\xi - \xi + (\xi - \xi')\right]^+}{1-\theta} \le (-\xi)^+ = \xi^-,$$

instead of (3.22), and it follows from (3.21) that

$$\left(\theta Y_t - Y'_t\right)^+ \le (1-\theta)\left(C\mathbb{E}\left[\psi\left(\xi^- + \int_0^T \alpha_s \mathrm{d}s,\ \mu_{k,0}(T);\ k\right)\middle| \mathcal{F}_t\right] + C\right), \quad t \in [0,T].$$

Thus, the desired conclusion follows in the limit as $\theta \to 1$.

Finally, in the same way as above, we can prove the desired conclusion under the conditions that the generator g' satisfies assumptions (EX2) and (UN3), and $\mathrm{d}\mathbb{P} \times \mathrm{d}t$-$a.e.$, $g(t, Y_t, Z_t) \le g'(t, Y_t, Z_t)$.

(ii) Case of $k \in [1/2, 1]$. Since both $\psi(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, \mu_{k,\varepsilon}(\cdot); k)$ and $\psi(|Y'_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, \mu_{k,\varepsilon}(\cdot); k)$ belong to class (D) for some $\varepsilon > 0$, we can follow closely the proof procedure of (i) with $\mu_{k,\varepsilon}(\cdot)$ instead of $\mu_{k,0}(\cdot)$ to obtain the desired assertion.

(iii) Case of $k = 2\lambda > 1$. We suppose that for some $\varepsilon > 0$ and $q \ge 1$, both processes $\psi(|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, 2q3^{k-1}\mu_{k,\varepsilon}(\cdot); k)$ and $\psi(|Y'_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s, 2q3^{k-1}\mu_{k,\varepsilon}(\cdot); k)$ belongs to class (D). Observe that for each $x, y, z \ge 0$, $c > 0$ and $k > 1$, we have

$$1 + cx + cy \le (1+c)(1+x)(1+y)$$

and

$$(x+y+z)^k \le 3^{k-1}\left(x^k + y^k + z^k\right).$$

It follows that for each $x, y \ge 0$, $\theta \in (0,1)$ and $k > 1$,

$$\left[\ln\left(1 + \frac{(x-\theta y)^+}{1-\theta}\right)\right]^k \le 3^{k-1}\left[\left(\ln\left(1 + \frac{1}{1-\theta}\right)\right)^k + (\ln(1+x))^k + (\ln(1+y))^k\right].$$

By virtue of the last inequality, Young's inequality, the definition of $\psi(\cdot,\cdot;k)$ with $k > 1$ and the above assumptions, it is not very hard to verify that the process

$$\psi\left(\left(\Delta^\theta U_t\right)^+ + \int_0^t \alpha_s \mathrm{d}s,\ \mu_{k,\varepsilon}(t);\ k\right), \quad t \in [0,T]$$

belongs to class (D) for each $\theta \in (0,1)$. The rest proof runs as in (i) or (ii). □

Proof of Theorem 2.13. Assume that $k = 1$ and there exists a random variable $X \in L^1$ such that $\mathbb{P}$-$a.s.$, for some $\varepsilon_0 > 0$,

$$\psi\left(|Y_t| + |Y'_t|, \mu_{k,\varepsilon_0}(t); k\right) \le \mathbb{E}[X|\mathcal{F}_t], \quad t \in [0,T]. \tag{3.23}$$

We only prove the case that the generator $g = g_1 + g_2$, g_1 satisfies (UN1) and (UN2), g_2 satisfies (EX2) and is convex in the state variables (y,z), and $\mathrm{d}\mathbb{P} \times \mathrm{d}t$-$a.e.$,

$$g(t, Y'_t, Z'_t) \le g'(t, Y'_t, Z'_t).$$

The other cases can be proved in the same way.

We first use the θ-technique. For each fixed $\theta \in (0,1)$, define

$$\Delta^\theta U_\cdot := \frac{Y_\cdot - \theta Y'_\cdot}{1-\theta} \quad \text{and} \quad \Delta^\theta V_\cdot := \frac{Z_\cdot - \theta Z'_\cdot}{1-\theta}. \tag{3.24}$$

Then the pair $(\Delta^\theta U_\cdot, \Delta^\theta V_\cdot)$ satisfies the following BSDE:

$$\Delta^\theta U_t = \Delta^\theta U_T + \int_t^T \frac{g(s,Y_s,Z_s) - \theta g'(s,Y'_s,Z'_s)}{1-\theta}\mathrm{d}s - \int_t^T \Delta^\theta V_s \cdot \mathrm{d}B_s, \qquad t \in [0,T]. \tag{3.25}$$

Denote $\Delta^\theta g_1(s) := g_1(s,Y_s,Z_s) - \theta g_1(s,Y'_s,Z'_s)$. Observe from the assumptions that $\mathrm{d}\mathbb{P} \times \mathrm{d}s$-$a.e.$,

$$\begin{aligned}
&\mathbf{1}_{\Delta^\theta U_s>0}(g(s,Y_s,Z_s) - \theta g'(s,Y'_s,Z'_s))\\
=&\mathbf{1}_{\Delta^\theta U_s>0}[g(s,Y_s,Z_s) - \theta g(s,Y'_s,Z'_s) + \theta(g(s,Y'_s,Z'_s) - g'(s,Y'_s,Z'_s))]\\
\le&\mathbf{1}_{\Delta^\theta U_s>0}\left\{\Delta^\theta g_1(s) + \left[g_2\left(s,\theta Y'_s + (1-\theta)\Delta^\theta U_s, \theta Z'_s + (1-\theta)\Delta^\theta V_s\right) - \theta g_2(s,Y'_s,Z'_s)\right]\right\}\\
\le&\mathbf{1}_{\Delta^\theta U_s>0}\left[\Delta^\theta g_1(s) + (1-\theta) g_2\left(s,\Delta^\theta U_s,\Delta^\theta V_s\right)\right]\\
\le&\mathbf{1}_{\Delta^\theta U_s>0}\Delta^\theta g_1(s) + (1-\theta)\Bigg[\alpha_s + \beta\left(\Delta^\theta U_s\right)^+\left(\ln(\left(\Delta^\theta U_s\right)^+)\right)^\delta \mathbf{1}_{(\Delta^\theta U_s)^+>1}\\
&+\ \gamma|\Delta^\theta V_s|\left|\ln|\Delta^\theta V_s|\right|^\lambda\Bigg]
\end{aligned}$$

and then

$$\begin{aligned}
&\mathbf{1}_{\Delta^\theta U_s>0}\frac{g(s,Y_s,Z_s) - \theta g'(s,Y'_s,Z'_s)}{1-\theta}\\
&\quad\le \frac{\mathbf{1}_{\Delta^\theta U_s>0}\Delta^\theta g_1(s)}{1-\theta} + \alpha_s + \beta\left(\Delta^\theta U_s\right)^+\left(\ln(\left(\Delta^\theta U_s\right)^+)\right)^\delta \mathbf{1}_{(\Delta^\theta U_s)^+>1}\\
&\qquad+\ \gamma|\Delta^\theta V_s|\left|\ln|\Delta^\theta V_s|\right|^\lambda.
\end{aligned} \tag{3.26}$$

Now, we pick $\varepsilon = \varepsilon_0/2$. Apply Itô-Tanaka's formula to the process $\varphi(s,\left(\Delta^\theta U_s\right)^+;k,\varepsilon)$, where the function $\varphi(\cdot,\cdot;k,\varepsilon)$ with $k=1$ is defined in (3.7), and in view of (3.25), (3.26) and Proposition 3.4, we use a similar argument to that in the proof of Proposition 3.6 to derive that for each $s \in [0,T]$,

$$\begin{aligned}
\mathrm{d}\varphi(s,\left(\Delta^\theta U_s\right)^+;k,\varepsilon) \ge& -\varphi_x(s,\left(\Delta^\theta U_s\right)^+;k,\varepsilon)\left(\frac{\mathbf{1}_{\Delta^\theta U_s>0}\Delta^\theta g_1(s)}{1-\theta} + \alpha_s(\omega)\right)\mathrm{d}s\\
&+\varphi_x(s,\left(\Delta^\theta U_s\right)^+;k,\varepsilon)\mathbf{1}_{\Delta^\theta U_s>0}\Delta^\theta V_s \cdot \mathrm{d}B_s.
\end{aligned} \tag{3.27}$$

For each $t \in [0,T]$ and each integer $n \ge 1$, define the following stopping time

$$\tau_n^t := \inf\left\{s \in [t,T] : |Y_s| + |Y'_s| + \int_t^s (|Z_r|^2 + |Z'_r|^2)\mathrm{d}r \ge n\right\} \wedge T.$$

By virtue of (3.27) and the definition of $\varphi(\cdot,\cdot;k,\varepsilon)$ with $k=1$, we deduce that for each $t\in[0,T]$,

$$
\begin{aligned}
&\left(\left(\Delta^\theta U_t\right)^+\right)^{1+\mu_{k,\varepsilon}(t)} \le e^{\nu_{k,\varepsilon}(t)}\left(k_\varepsilon+\left(\Delta^\theta U_t\right)^+\right)^{1+\mu_{k,\varepsilon}(t)}\\
\le\ & e^{\nu_{k,\varepsilon}(T)}\left(k_\varepsilon+\left(\Delta^\theta U_{\tau_n^t}\right)^+\right)^{1+\mu_{k,\varepsilon}(\tau_n^t)}\\
&+\int_t^{\tau_n^t} e^{\nu_{k,\varepsilon}(s)}\left(1+\mu_{k,\varepsilon}(s)\right)\left(k_\varepsilon+\left(\Delta^\theta U_s\right)^+\right)^{\mu_{k,\varepsilon}(s)}\left(\frac{\Delta^\theta g_1(s)}{1-\theta}\mathbf{1}_{\Delta^\theta U_s>0}+\alpha_s(\omega)\right)\mathrm{d}s\\
&-\int_t^{\tau_n^t} e^{\nu_{k,\varepsilon}(s)}\left(1+\mu_{k,\varepsilon}(s)\right)\left(k_\varepsilon+\left(\Delta^\theta U_s\right)^+\right)^{\mu_{k,\varepsilon}(s)}\mathbf{1}_{\Delta^\theta U_s>0}\Delta^\theta V_s\cdot\mathrm{d}B_s,
\end{aligned}
$$

and then in view of (3.24), for each $\theta\in(0,1)$ and $n\ge 1$,

$$
\begin{aligned}
&\left((Y_t-\theta Y'_t)^+\right)^{1+\mu_{k,\varepsilon}(t)}\\
\le\ & K_T\left((1-\theta)k_\varepsilon+(Y_{\tau_n^t}-\theta Y'_{\tau_n^t})^+\right)^{1+\mu_{k,\varepsilon}(\tau_n^t)}\\
&+\int_t^{\tau_n^t} K_s\left((1-\theta)k_\varepsilon+(Y_s-\theta Y'_s)^+\right)^{\mu_{k,\varepsilon}(s)}\left[\Delta^\theta g_1(s)\mathbf{1}_{\Delta^\theta U_s>0}+(1-\theta)\alpha_s(\omega)\right]\mathrm{d}s\\
&-\int_t^{\tau_n^t} K_s\left((1-\theta)k_\varepsilon+(Y_s-\theta Y'_s)^+\right)^{\mu_{k,\varepsilon}(s)}\mathbf{1}_{\Delta^\theta U_s>0}(Z_s-\theta Z'_s)\cdot\mathrm{d}B_s,\quad t\in[0,T],
\end{aligned}
\tag{3.28}
$$

where

$$
K_s:=e^{\nu_{k,\varepsilon}(s)}\left(1+\mu_{k,\varepsilon}(s)\right),\quad s\in[0,T].
$$

Moreover, define $\hat U_\cdot:=Y_\cdot-Y'_\cdot$ and $\hat V_\cdot:=Z_\cdot-Z'_\cdot$. In view of the definition of $\Delta^\theta g_1(s)$ and assumptions (UN1) and (UN2) of g_1, sending $\theta\to1$ in (3.28) implies that for each $n\ge1$,

$$
\begin{aligned}
\left(\hat U_t^+\right)^{1+\mu_{k,\varepsilon}(t)} \le\ & K_T\left(\hat U_{\tau_n^t}^+\right)^{1+\mu_{k,\varepsilon}(\tau_n^t)}+\int_t^{\tau_n^t}K_s\left(\hat U_s^+\right)^{\mu_{k,\varepsilon}(s)}\left(\rho(\hat U_s^+)+\kappa(|\hat V_s|)\right)\mathrm{d}s\\
&-\int_t^{\tau_n^t}K_s\left(\hat U_s^+\right)^{\mu_{k,\varepsilon}(s)}\hat V_s\cdot\mathrm{d}B_s,\quad t\in[0,T].
\end{aligned}
\tag{3.29}
$$

In view of Remark 2.2, since $\kappa(\cdot)$ is a continuous function of linear growth with $\kappa(0)=0$, it follows from the proof of Theorem 1 in [8] that for each $m\ge1$,

$$
\forall x\in\mathbb{R}_+,\quad \kappa(x)\le(m+2A)x+\kappa\left(\frac{2A}{m+2A}\right).\tag{3.30}
$$

Let $\mathbb{P}_m$ be the probability on $(\Omega,\mathcal{F})$ which is equivalent to $\mathbb{P}$ defined by

$$
\frac{\mathrm{d}\mathbb{P}_m}{\mathrm{d}\mathbb{P}}:=\exp\left\{(m+2A)\int_0^T\frac{\hat V_s}{|\hat V_s|}\mathbf{1}_{|\hat V_s|>0}\cdot\mathrm{d}B_s+\frac12(m+2A)^2\int_0^T\mathbf{1}_{|\hat V_s|>0}\mathrm{d}s\right\}.
$$

Note that $\mathrm{d}\mathbb{P}_m/\mathrm{d}\mathbb{P}$ has moments of all orders. Let $\mathbb{E}_m[\eta|\mathcal{F}_t]$ represent the conditional mathematical expectation of the random variable η with respect to $\mathcal{F}_t$ under the probability measure $\mathbb{P}_m$.

Then, in view of (3.29) and (3.30), by Girsanov's theorem we deduce that for each $n, m \geq 1$,

$$\begin{aligned}\left(\hat{U}_t^+\right)^{1+\mu_{k,\varepsilon}(t)} &\leq \kappa\left(\frac{2A}{m+2A}\right) K_T T + K_T \mathbb{E}_m\left[\left(\hat{U}_{\tau_n^t}^+\right)^{1+\mu_{k,\varepsilon}(\tau_n^t)}\middle|\mathcal{F}_t\right] \\ &\quad + K_T \mathbb{E}_m\left[\int_t^{\tau_n^t}\left(\hat{U}_s^+\right)^{\mu_{k,\varepsilon}(s)}\rho(\hat{U}_s^+)\mathrm{d}s\middle|\mathcal{F}_t\right], \quad t\in[0,T].\end{aligned} \tag{3.31}$$

On the other hand, denote

$$p := \inf_{s\in[0,T]}\frac{1+\mu_{k,\varepsilon_0}(s)}{1+\mu_{k,\varepsilon}(s)} \quad \text{and} \quad P_t := \left(\hat{U}_t^+\right)^{1+\mu_{k,\varepsilon}(t)}, \quad t\in[0,T].$$

Since $\varepsilon = \varepsilon_0/2$, from the definition of $\mu_{k,\varepsilon}(\cdot)$ with $k = 1$ it is not hard to conclude that $p > 1$. We pick a $q \in (1, p)$. It follows from [16, Lemma 6.1], (2.4) and (3.23) that

$$\begin{aligned}\mathbb{E}\left[\sup_{t\in[0,T]}|P_t|^q\right] &\leq \mathbb{E}\left[\sup_{t\in[0,T]}\left[\left(|Y_t|+|Y_t'|\right)^{1+\mu_{k,\varepsilon_0}(t)}\right]^{\frac{q}{p}}\right] \\ &\leq \mathbb{E}\left[\sup\nolimits_{t\in[0,T]}\left(\psi\left(|Y_t|+|Y_t'|,\mu_{k,\varepsilon_0}(t);k\right)\right)^{\frac{q}{p}}\right] \\ &\leq \mathbb{E}\left[\left(\sup_{t\in[0,T]}\mathbb{E}[X|\mathcal{F}_t]\right)^{\frac{q}{p}}\right] \leq \frac{p}{p-q}\left(\mathbb{E}[X]\right)^{\frac{q}{p}} < +\infty,\end{aligned}$$

which means that $P_\cdot \in \mathcal{S}^q$, and then, by virtue of Hölder's inequality,

$$\forall m\geq 1, \quad \mathbb{E}_m\left[\sup_{t\in[0,T]}|P_t|\right] = \mathbb{E}\left[\left(\sup_{t\in[0,T]}|P_t|\right)\frac{\mathrm{d}\mathbb{P}_m}{\mathrm{d}\mathbb{P}}\right] < +\infty. \tag{3.32}$$

Thus, in view of $\hat{U}_T^+ = 0$ and the fact that $\rho(\cdot)$ is of linear growth, we can send n to infinity in (3.31) to obtain that for each $m \geq 1$,

$$P_t \leq \kappa\left(\frac{2A}{m+2A}\right)K_T T + K_T\mathbb{E}_m\left[\int_t^T P_s^{\frac{\mu_{k,\varepsilon}(s)}{1+\mu_{k,\varepsilon}(s)}}\rho\left(P_s^{\frac{1}{1+\mu_{k,\varepsilon}(s)}}\right)\mathrm{d}s\middle|\mathcal{F}_t\right], \quad t\in[0,T]. \tag{3.33}$$

Finally, define the function

$$\bar{\rho}(x) := \sup_{t\in[0,T]}\left[x^{\frac{\mu_{k,\varepsilon}(t)}{1+\mu_{k,\varepsilon}(t)}}\rho\left(x^{\frac{1}{1+\mu_{k,\varepsilon}(t)}}\right)\right], \quad x\in\mathbb{R}_+.$$

It is clear that $\bar{\rho}(0) = 0$. Since $\rho(x)$ is a nondecreasing concave function on $\mathbb{R}_+$ with $\rho(0) = 0$, it follows from [19, Lemma 6.1] that $\rho(x)/x,\ x > 0$ is a non-increasing function, and then,

$$\bar{\rho}(x) = x\sup_{t\in[0,T]}\left[\frac{\rho\left(x^{\frac{1}{1+\mu_{k,\varepsilon}(t)}}\right)}{x^{\frac{1}{1+\mu_{k,\varepsilon}(t)}}}\right] = x\,\frac{\rho\left(x^{\frac{1}{1+\varepsilon}}\right)}{x^{\frac{1}{1+\varepsilon}}} = x^{\frac{\varepsilon}{1+\varepsilon}}\rho\left(x^{\frac{1}{1+\varepsilon}}\right), \quad 0<x\leq 1,$$

and

$$\bar{\rho}(x) = x\sup_{t\in[0,T]}\left[\frac{\rho\left(x^{\frac{1}{1+\mu_{k,\varepsilon}(t)}}\right)}{x^{\frac{1}{1+\mu_{k,\varepsilon}(t)}}}\right] = x\,\frac{\rho\left(x^{\frac{1}{1+\mu_{k,\varepsilon}(T)}}\right)}{x^{\frac{1}{1+\mu_{k,\varepsilon}(T)}}} = x^{\frac{\mu_{k,\varepsilon}(T)}{1+\mu_{k,\varepsilon}(T)}}\rho\left(x^{\frac{1}{1+\mu_{k,\varepsilon}(T)}}\right), \quad x\geq 1.$$

Then, $\bar{\rho}(\cdot)$ is well defined on $\mathbb{R}_+$, nondecreasing, continuous and of linear growth, and

$$\int_{0^+} \frac{\mathrm{d}x}{\bar{\rho}(x)} = \int_{0^+} \frac{\mathrm{d}x}{x^{\frac{\varepsilon}{1+\varepsilon}} \rho\left(x^{\frac{1}{1+\varepsilon}}\right)} = (1+\varepsilon) \int_{0^+} \frac{\mathrm{d}u}{\rho(u)} = +\infty. \tag{3.34}$$

Now, coming back to (3.33), we have that for each $m \geq 1$,

$$P_t \leq \kappa \left(\frac{2A}{m+2A}\right) K_T T + K_T \mathbb{E}_m \left[\int_t^T \bar{\rho}(P_s) \mathrm{d}s \middle| \mathcal{F}_t\right], \quad t \in [0, T]. \tag{3.35}$$

Then, in view of (3.32), (3.34), (3.35) and the fact that

$$\lim_{m\to\infty} \kappa \left(\frac{2A}{m+2A}\right) = 0,$$

applying [22, Lemma 2.1] implies that $\mathbb{P}$-$a.s.$, for each $t \in [0, T]$, $P_t = 0$, that is, $Y_t \leq Y'_t$.

The proof is complete. □

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Proof of technical propositions

In this section, we will give the proofs of several technical propositions in the last section.

Proof of Proposition 3.1. We use the localization procedure to construct the desired solution. For each integer $m \geq 1$, define the following stopping time:

$$\sigma_m := \inf\left\{t \in [0, T] : X_t + \int_0^t f_s \mathrm{d}s \geq m\right\} \wedge T$$

with the convention that $\inf \emptyset = +\infty$. Then $(Y_m^{n,p}(t), Z_m^{n,p}(t)) := (Y_{t\wedge\sigma_m}^{n,p}, Z_t^{n,p} \mathbf{1}_{t\leq\sigma_m})$ solves the BSDE:

$$Y_m^{n,p}(t) = Y_{\sigma_m}^{n,p} + \int_t^T \mathbf{1}_{s\leq\sigma_m} g^{n,p}(s, Y_m^{n,p}(s), Z_m^{n,p}(s)) \mathrm{d}s - \int_t^T Z_m^{n,p}(s) \cdot \mathrm{d}B_s, \quad t \in [0, T].$$

Note that for fixed $m \geq 1$, $Y_m^{n,p}(\cdot)$ is nondecreasing in n, non-increasing in p and bounded by m, and that $\mathrm{d}\mathbb{P} \times \mathrm{d}t$-$a.e.$, the sequence $(g^{n,p})_{n,p}$ converges locally uniformly in (y, z) to the generator g as $n, p \to \infty$. Furthermore, in view of the facts that $|g^{n,p}| \leq |g|$ and g satisfies (EX1), it follows that $\mathrm{d}\mathbb{P} \times \mathrm{d}s$-$a.e.$,

$$\forall (y, z) \in [-m, m] \times \mathbb{R}^d, \quad \sup_{n,p\geq 1} \left(\mathbf{1}_{s\leq\sigma_m} \left|g^{n,p}(s, y, z)\right|\right) \leq \mathbf{1}_{s\leq\sigma_m} f_s + H(m) + c|z|^2,$$

with $\mathbb{P}$-$a.s.$, $\int_0^T \mathbf{1}_{s\leq\sigma_m} f_s \mathrm{d}s \leq m$. Thus, we can apply the stability property for bounded solutions of BSDEs (see e.g. Proposition 3.1 in [42]). Setting $Y_m(t) := \inf_p \sup_n Y_{t\wedge\sigma_m}^{n,p}$, then $Y_m(\cdot)$ is continuous and there exists a process $Z_m(\cdot)$ such that

$$\lim_{n,p\to\infty} Z_t^{n,p} \mathbf{1}_{t\leq\sigma_m} = Z_m(t) \text{ in } \mathcal{M}^2$$

and

$$Y_m(t) = \inf_p \sup_n Y^{n,p}_{\sigma_m} + \int_t^T \mathbf{1}_{s\le\sigma_m} g(s, Y_m(s), Z_m(s))\mathrm{d}s - \int_t^T Z_m(s)\cdot \mathrm{d}B_s, \quad t\in[0,T].$$

Finally, in view of the assumptions of Proposition 3.1, the stability of stopping time σ_m and the fact that $\mathrm{d}\mathbb{P}\times\mathrm{d}t$-$a.e.$, for each $m\ge 1$,

$$Y_{m+1}(t\wedge\sigma_m) = Y_m(t\wedge\sigma_m) = \inf_{p\ge 1}\sup_{n\ge 1} Y^{n,p}_{t\wedge\sigma_m}$$

and

$$Z_{m+1}\mathbf{1}_{t\le\sigma_m} = Z_m\mathbf{1}_{t\le\sigma_m} = \lim_{n,p\to\infty} Z^{n,p}_t\mathbf{1}_{t\le\sigma_m},$$

the conclusion of Proposition 3.1 follows immediately by picking

$$Z_t := \sum_{m=1}^{+\infty} Z_m(t)\mathbf{1}_{t\in[\sigma_{m-1},\sigma_m)}, \qquad t\in[0,T]$$

with $\sigma_0 := 0$. The proof is complete. □

Proof of Proposition 3.2. Note first that (3.5) can be rewritten as

$$2\frac{y}{x}\left|\ln\frac{y}{x}+\ln x\right|^{\lambda} \le k4^{(\lambda-1)^+}|\ln x|^{2\lambda} + C_{k,\lambda} + \left(\frac{y}{x}\right)^2.$$

Set $a := y/x > 0$. Then, to prove (3.5), it is equivalent to prove that

$$\forall a,x>0,\quad 2a|\ln a+\ln x|^{\lambda} \le p4^{(\lambda-1)^+}|\ln x|^{2\lambda} + a^2 + C_{p,\lambda}. \tag{A.1}$$

Now, we let $p>1$ and prove (A.1). By virtue of Young's inequality, observe that for each $a,x>0$,

$$|\ln a+\ln x|^{\lambda} \le 2^{(\lambda-1)^+}|\ln a|^{\lambda} + 2^{(\lambda-1)^+}|\ln x|^{\lambda} \tag{A.2}$$

and

$$2a\left(2^{(\lambda-1)^+}|\ln x|^{\lambda}\right) \le \frac{1}{p}a^2 + p4^{(\lambda-1)^+}|\ln x|^{2\lambda}. \tag{A.3}$$

And, note that the function

$$f(a;p,\lambda) := 2a\left(2^{(\lambda-1)^+}|\ln a|^{\lambda}\right) - \left(1-\frac{1}{p}\right)a^2$$

is continuous on $a\in(0,+\infty)$, and respectively tends to 0 and $-\infty$ as a tends to 0^+ and $+\infty$, and then there must exist a constant $C_{p,\lambda}>0$ depending only on (p,λ) such that

$$\forall a>0,\quad f(a;p,\lambda)\le C_{p,\lambda}. \tag{A.4}$$

Combining (A.2), (A.3) and (A.4) yields that

$$\begin{aligned}
&\quad 2a|\ln a+\ln x|^{\lambda} - p4^{(\lambda-1)^+}|\ln x|^{2\lambda} - a^2\\
&\le 2a\left(2^{(\lambda-1)^+}|\ln a|^{\lambda}\right) - \left(1-\frac{1}{p}\right)a^2 + 2a\left(2^{(\lambda-1)^+}|\ln x|^{\lambda}\right) - p4^{(\lambda-1)^+}|\ln x|^{2\lambda} - \frac{1}{p}a^2\\
&\le f(a;p,\lambda)\le C_{p,\lambda},\quad \forall a,x>0.
\end{aligned}$$

This is just (A.1). Thus, we have proved (3.5) for each $p>1$.

Next, we show the second part of this proposition. In fact, it suffices to prove that (A.1) does not hold for $p = 1/4^{(\lambda-1)^+}$, i.e., there does not exist a positive constant $C_\lambda > 0$ such that

$$\forall a, x > 0, \quad 2a|\ln a + \ln x|^\lambda \le |\ln x|^{2\lambda} + a^2 + C_\lambda. \tag{A.5}$$

Let $\bar{a}(x) := |\ln x|^\lambda$. Then $2\bar{a}(x)|\ln x|^\lambda = |\ln x|^{2\lambda} + \bar{a}^2(x)$. Thus, if (A.5) holds, then we have

$$g(x;\lambda) := 2\bar{a}(x)|\ln \bar{a}(x) + \ln x|^\lambda - 2\bar{a}(x)|\ln x|^\lambda \le C_\lambda, \quad x > 0,$$

which is impossible since $g(x;\lambda)$ tends to positive infinity as $x \to +\infty$. The proof is then complete. □

Proof of Proposition 3.4. We distinguish four different cases: $k = 1/2$, $k \in (1/2, 1)$, $k = 1$ and $k > 1$ to prove this proposition respectively.

(i) Case $k = 1/2$. We first consider the special case of $\lambda = 0$ and $k = \delta = 1/2$. Choose the following function

$$\phi(s,x) := (x+e)\exp\left(\mu_s\sqrt{\ln(x+e)} + \nu_s\right) > 0, \quad (s,x) \in [0,T] \times \mathbb{R}_+ \tag{A.6}$$

to explicitly solve the inequality (3.4) with $\delta = 1/2$, where $\mu_s, \nu_s : [0,T] \to \mathbb{R}_+$ are two strictly increasing and continuous functions with zero only at the origin, and are continuously differential on the time interval $(0,T]$. For each $(s,x) \in (0,T] \times \mathbb{R}_+$, a simple computation gives

$$\phi_x(s,x) = \phi(s,x)\frac{\mu_s + 2\sqrt{\ln(x+e)}}{2(x+e)\sqrt{\ln(x+e)}} > 0, \tag{A.7}$$

$$\phi_{xx}(s,x) = \phi(s,x)\frac{\mu_s\left(2\ln(x+e) + \mu_s\sqrt{\ln(x+e)} - 1\right)}{4(x+e)^2\left(\sqrt{\ln(x+e)}\right)^3} > 0 \tag{A.8}$$

and

$$\phi_s(s,x) = \phi(s,x)\left(\mu_s'\sqrt{\ln(x+e)} + \nu_s'\right) > 0. \tag{A.9}$$

Note that

$$\forall x \in \mathbb{R}_+, \quad x\sqrt{\ln x}\mathbf{1}_{x>1} \le (x+e)\sqrt{\ln(x+e)}. \tag{A.10}$$

Denote $v = \sqrt{\ln(x+e)}$. Substituting (A.7), (A.8), (A.9) and (A.10) into the left side of (3.4) with $\delta = 1/2$ yields that

$$\begin{aligned}
&-\beta\phi_x(s,x)x\sqrt{\ln x}\mathbf{1}_{x>1} - \frac{\gamma^2}{2}\frac{\phi_x^2(s,x)}{\phi_{xx}(s,x)} + \phi_s(s,x)\\
&\ge -\beta\phi(s,x)(\mu_s+v) - \frac{\gamma^2}{2}\phi(s,x)\frac{v(\mu_s+2v)^2}{\mu_s(2v^2+\mu_s v-1)}\\
&\quad + \phi(s,x)(\mu_s' v + \nu_s'), \quad (s,x) \in (0,T] \times \mathbb{R}_+.
\end{aligned}$$

Furthermore, in view of the fact of $v \ge 1$, we know that

$$\begin{aligned}
\frac{v\,(\mu_s+2v)^2}{2v^2+\mu_s v-1} &= \frac{(2v+\mu_s)(2v^2+\mu_s v-1) + (2v+\mu_s)}{2v^2+\mu_s v-1}\\
&\le 2v + \mu_s + \frac{2v+\mu_s}{\frac{1}{2}(2v^2+\mu_s v)} = 2v + \mu_s + \frac{2}{v} \le 2v + \mu_s + 2,
\end{aligned}$$

and then

$$\begin{aligned}&-\beta\phi_x(s,x)x\sqrt{\ln x}\mathbf{1}_{x>1}-\frac{\gamma^2}{2}\frac{\phi_x^2(s,x)}{\phi_{xx}(s,x)}+\phi_s(s,x)\\&\geq\phi(s,x)\left\{\left(\mu_s'-\frac{\gamma^2}{\mu_s}-\beta\right)v+\left[\nu_s'-\beta\mu_s-\frac{\gamma^2}{2}\left(1+\frac{2}{\mu_s}\right)\right]\right\},\qquad(s,x)\in(0,T]\times\mathbb{R}_+.\end{aligned}$$

Thus, (3.4) with $\delta=1/2$ holds for the function $\phi(\cdot,\cdot)$ if the functions $\mu_s,\nu_s\in[0,T]$ satisfies

$$\mu_s'=\frac{\gamma^2}{\mu_s}+\beta,\quad s\in(0,T];\quad \mu_0=0 \tag{A.11}$$

and

$$\nu_s'=\beta\mu_s+\frac{\gamma^2}{2}\left(1+\frac{2}{\mu_s}\right),\quad s\in(0,T]. \tag{A.12}$$

It is clear that $\mu_{k,0}(\cdot)$ with $k=\delta=1/2$ is just the unique strictly increasing and continuous solution of ODE (A.11). And, by integrating on both sides of (A.11), we have

$$\mu_{k,0}(T)-\mu_{k,0}(t)=\gamma^2\int_t^T\frac{1}{\mu_{k,0}(s)}\mathrm{d}s+\beta(T-t),\quad t\in(0,T].$$

Letting $t\to0^+$ in the last equation yields that the integral of $1/\mu_{k,0}(s)$ on $[0,T]$ is well defined, and then (A.12) admits a unique solution valued zero at the origin, denoted by $\nu_{k,0}(\cdot)$, i.e.,

$$\nu_{k,0}(s)=\int_0^s\left[\beta\mu_{k,0}(r)+\frac{\gamma^2}{2}\left(1+\frac{2}{\mu_{k,0}(r)}\right)\right]\mathrm{d}r,\quad s\in[0,T]. \tag{A.13}$$

Note that when $k=1/2$ and $\varepsilon=0$, the function $\varphi(\cdot,\cdot;k,\varepsilon)$ defined in (3.7) is just the function $\phi(\cdot,\cdot)$ defined in (A.6) with $\mu(\cdot)=\mu_{k,0}(\cdot)$ and $\nu(\cdot)=\nu_{k,0}(\cdot)$. Consequently, the desired conclusion is true for $\varepsilon=0$ in the case of $k=\delta=1/2$.

Finally, by virtue of the definition of k_ε, we have for each $\delta\in[0,1/2)$ and each $\varepsilon>0$,

$$\forall x\in\mathbb{R}_+,\quad x(\ln x)^\delta\mathbf{1}_{x>1}\leq\varepsilon(x+k_\varepsilon)\sqrt{\ln(x+k_\varepsilon)}.$$

By respectively replacing e with k_ε and (A.10) with the last inequality in the above computation, and in view of the definitions of $\varphi(\cdot,\cdot;k,\varepsilon)$ and $\mu_{k,\varepsilon}(\cdot)$ with $k=1/2$ and $\varepsilon>0$, we can deduce that in the case that $k=1/2$ and $\delta\in[0,1/2)$, the desired conclusion is also true for $\varepsilon>0$.

(ii) Case $k\in(1/2,1)$. We first consider the special case of $k=\delta=\lambda+1/2\in(1/2,1)$. Let $\varepsilon>0$ and choose the following function

$$\phi(s,x;\varepsilon):=(x+k_\varepsilon)\exp\left(\mu_s\left(\ln\left(x+k_\varepsilon\right)\right)^{\lambda+1/2}+\nu_s\right),\quad(s,x)\in[0,T]\times\mathbb{R}_+ \tag{A.14}$$

to explicitly solve the inequality (3.6) with $\delta=\lambda+1/2$, where $\mu_s,\nu_s:[0,T]\to\mathbb{R}_+$ be two increasing and continuously differential functions with $\mu_0=\varepsilon$, $\mu_T=\mu_{k,\varepsilon}(T)$ and $\nu_0=0$. For each $(s,x)\in[0,T]\times\mathbb{R}_+$, a simple computation gives

$$\phi_x(s,x;\varepsilon)=\phi(s,x;\varepsilon)\frac{\left(\lambda+\frac12\right)\mu_s+(\ln(x+k_\varepsilon))^{\frac12-\lambda}}{(x+k_\varepsilon)\left(\ln(x+k_\varepsilon)\right)^{\frac12-\lambda}}>0, \tag{A.15}$$

$$\phi_{xx}(s,x;\varepsilon)=\phi(s,x;\varepsilon)\frac{\left(\lambda+\frac12\right)\mu_s\left[\ln(x+k_\varepsilon)+\left(\lambda+\frac12\right)\mu_s\left(\ln(x+k_\varepsilon)\right)^{\lambda+\frac12}-\left(\frac12-\lambda\right)\right]}{(x+k_\varepsilon)^2\left(\ln(x+k_\varepsilon)\right)^{\frac32-\lambda}}>0$$

(A.16)

and

$$\phi_s(s,x;\varepsilon)=\phi(s,x;\varepsilon)\left(\mu_s'\left(\ln(x+k_\varepsilon)\right)^{\lambda+\frac12}+\nu_s'\right)>0. \tag{A.17}$$

Furthermore, from the definition of k_ε it can be directly verified that for each $(s,x)\in[0,T]\times\mathbb{R}_+$,

$$\begin{aligned}&\ln(x+k_\varepsilon)+\left(\lambda+\frac12\right)\mu_s\left(\ln(x+k_\varepsilon)\right)^{\lambda+\frac12}-\left(\frac12-\lambda\right)\\ \geq\ &\frac{1}{1+\varepsilon}\left[\ln(x+k_\varepsilon)+\left(\lambda+\frac12\right)\mu_s\left(\ln(x+k_\varepsilon)\right)^{\lambda+\frac12}\right]\end{aligned} \tag{A.18}$$

and

$$\left(\ln(x+k_\varepsilon)\right)^{\frac12-\lambda}\leq\sqrt{\ln(x+k_\varepsilon)}\leq(x+k_\varepsilon)^{\varepsilon}. \tag{A.19}$$

It follows from (A.16) and (A.18) that

$$\phi_{xx}(s,x;\varepsilon)\geq\phi(s,x;\varepsilon)\frac{\left(\lambda+\frac12\right)\mu_s\left[\left(\lambda+\frac12\right)\mu_s+\left(\ln(x+k_\varepsilon)\right)^{\frac12-\lambda}\right]}{(1+\varepsilon)(x+k_\varepsilon)^2\left(\ln(x+k_\varepsilon)\right)^{1-2\lambda}},\quad (s,x)\in[0,T]\times\mathbb{R}_+.$$

Then, for each $(s,x)\in[0,T]\times\mathbb{R}_+$, in view of (A.15), we have

$$\frac{\gamma^2}{2}\frac{(\phi_x(s,x;\varepsilon))^2}{\phi_{xx}(s,x;\varepsilon)}\leq\frac{\gamma^2(1+\varepsilon)}{2}\phi(s,x;\varepsilon)\left(1+\frac{\left(\ln(x+k_\varepsilon)\right)^{\frac12-\lambda}}{\left(\lambda+\frac12\right)\mu_s}\right) \tag{A.20}$$

and, in view of the definition of k_ε, (A.16) and (A.19),

$$\begin{cases}\dfrac{\gamma\phi_x(s,x;\varepsilon)}{\phi_{xx}(s,x;\varepsilon)}\geq\dfrac{\gamma}{\left(\lambda+\frac12\right)\mu_s}(x+k_\varepsilon)\left(\ln(x+k_\varepsilon)\right)^{\frac12-\lambda}\geq\dfrac{\gamma}{\mu_T}k_\varepsilon=\dfrac{\gamma}{\mu_{k,\varepsilon}(T)}k_\varepsilon\geq1;\\[2ex] \dfrac{\gamma\phi_x(s,x;\varepsilon)}{\phi_{xx}(s,x;\varepsilon)}\leq\dfrac{\gamma(1+\varepsilon)}{\left(\lambda+\frac12\right)\mu_s}(x+k_\varepsilon)\left(\ln(x+k_\varepsilon)\right)^{\frac12-\lambda}\leq\dfrac{2\gamma(1+\varepsilon)}{\varepsilon}(x+k_\varepsilon)^{1+\varepsilon},\end{cases}$$

which yields the following

$$\left|\ln\frac{\gamma\phi_x(s,x;\varepsilon)}{\phi_{xx}(s,x;\varepsilon)}\right|^{2\lambda}\leq\left|\ln\frac{2\gamma(1+\varepsilon)}{\varepsilon}\right|^{2\lambda}+(1+\varepsilon)^{2\lambda}\left(\ln(x+k_\varepsilon)\right)^{2\lambda}. \tag{A.21}$$

In the sequel, observe that

$$\forall x\in\mathbb{R}_+,\quad x(\ln x)^{\lambda+\frac12}\mathbf{1}_{x>1}\leq(x+k_\varepsilon)\left(\ln(x+k_\varepsilon)\right)^{\lambda+\frac12}.$$

We substitute (A.15), (A.20), (A.17) and (A.21) into the left side of (3.6) with $\delta=1/2$ and $\phi(s,x;\varepsilon)$ instead of $\phi(s,x)$ to get

$$\begin{aligned}&-\beta\phi_x(s,x;\varepsilon)x(\ln x)^{\lambda+\frac12}\mathbf{1}_{x>1}-\frac{\gamma^2}{2}\frac{(\phi_x(s,x;\varepsilon))^2}{\phi_{xx}(s,x;\varepsilon)}\left[(1+\varepsilon)\left|\ln\frac{\gamma\phi_x(s,x;\varepsilon)}{\phi_{xx}(s,x;\varepsilon)}\right|^{2\lambda}+C_{\varepsilon,\lambda}\right]\\ &+\phi_s(s,x;\varepsilon)\\ \geq\ &-\beta\phi(s,x;\varepsilon)\left[\left(\lambda+\frac12\right)\mu_s\left(\ln(x+k_\varepsilon)\right)^{2\lambda}+\left(\ln(x+k_\varepsilon)\right)^{\lambda+\frac12}\right]\\ &-\frac{\gamma^2(1+\varepsilon)}{2}\phi(s,x;\varepsilon)\left(1+\frac{\left(\ln(x+k_\varepsilon)\right)^{\frac12-\lambda}}{\left(\lambda+\frac12\right)\mu_s}\right)\left[(1+\varepsilon)^{2\lambda+1}\left(\ln(x+k_\varepsilon)\right)^{2\lambda}+\bar{C}_{\gamma,\lambda,\varepsilon}\right]\\ &+\phi(s,x;\varepsilon)\left(\mu_s'\left(\ln(x+k_\varepsilon)\right)^{\lambda+\frac12}+\nu_s'\right),\qquad(s,x)\in[0,T]\times\mathbb{R}_+,\end{aligned}$$

where

$$\bar{C}_{\gamma,\lambda,\varepsilon} := (1+\varepsilon)\left|\ln\frac{2\gamma(1+\varepsilon)}{\varepsilon}\right|^{2\lambda} + C_{\varepsilon,\lambda}.$$

Furthermore, in view of the facts that $\mu_\cdot \geq \varepsilon$ and $\lambda \in (0, 1/2)$ together with the definition of k_ε, using Young's inequality we see that there is a constant $\tilde{C}_{\gamma,\lambda,\varepsilon} > 0$ depending only on $(\gamma, \lambda, \varepsilon, \bar{C}_{\gamma,\lambda,\varepsilon})$ such that for each $(s, x) \in [0, T] \times \mathbb{R}_+$,

$$(\ln(x+k_\varepsilon))^{2\lambda} \leq \varepsilon\,(\ln(x+k_\varepsilon))^{\lambda+\frac{1}{2}}$$

and

$$\begin{aligned}
&\frac{\gamma^2(1+\varepsilon)}{2}\left(1+\frac{(\ln(x+k_\varepsilon))^{\frac{1}{2}-\lambda}}{\left(\lambda+\frac{1}{2}\right)\mu_s}\right)\left[(1+\varepsilon)^{2\lambda+1}(\ln(x+k_\varepsilon))^{2\lambda}+\bar{C}_{\gamma,\lambda,\varepsilon}\right]\\
&\leq \left(\frac{\gamma^2(1+\varepsilon)^{2\lambda+2}}{(2\lambda+1)\mu_s}+\varepsilon\right)(\ln(x+k_\varepsilon))^{\lambda+\frac{1}{2}}+\tilde{C}_{\gamma,\lambda,\varepsilon}.
\end{aligned}$$

Then, with the notation $v := (\ln(x+k_\varepsilon))^{\lambda+\frac{1}{2}}$ we have for each $(s, x) \in [0, T] \times \mathbb{R}_+$,

$$\begin{aligned}
&-\beta\phi_x(s,x;\varepsilon)x(\ln x)^{\lambda+\frac{1}{2}}\mathbf{1}_{x>1}-\frac{\gamma^2}{2}\frac{(\phi_x(s,x;\varepsilon))^2}{\phi_{xx}(s,x;\varepsilon)}\left[(1+\varepsilon)\left|\ln\frac{\gamma\phi_x(s,x;\varepsilon)}{\phi_{xx}(s,x;\varepsilon)}\right|^{2\lambda}+C_{\varepsilon,\lambda}\right]\\
&+\phi_s(s,x;\varepsilon)\\
&\geq \phi(s,x;\varepsilon)\left[\left(-\beta\left(\lambda+\frac{1}{2}\right)\varepsilon\mu_s-\beta-\frac{\gamma^2(1+\varepsilon)^{2\lambda+2}}{(2\lambda+1)\mu_s}-\varepsilon+\mu_s'\right)v+\left(-\tilde{C}_{\gamma,\lambda,\varepsilon}+\nu_s'\right)\right].
\end{aligned}$$

Thus, (3.6) with $\delta = 1/2$ and $\phi(s, x; \varepsilon)$ instead of $\phi(s, x)$ holds if the functions $\mu_s, \nu_s \in [0, T]$ satisfies

$$\mu_s' = \varepsilon\beta\left(\lambda+\frac{1}{2}\right)\mu_s+\frac{\gamma^2(1+\varepsilon)^{2\lambda+2}}{2\lambda+1}\frac{1}{\mu_s}+\varepsilon+\beta, \quad s\in[0,T] \tag{A.22}$$

and

$$\nu_s = \tilde{C}_{\gamma,\lambda,\varepsilon}s, \quad s\in[0,T]. \tag{A.23}$$

It is clear that $\mu_{k,\varepsilon}(\cdot)$ with $k = \delta = \lambda + 1/2 \in (1/2, 1)$ is just the unique strictly increasing and continuous solution of ODE (A.22) with ε at the origin. And, we let $\nu_{k,\varepsilon}(\cdot) := \nu(\cdot)$. Note that when $k = \delta = \lambda + 1/2 \in (1/2, 1)$, the function $\varphi(\cdot, \cdot; k, \varepsilon)$ defined in (3.7) is just the function $\phi(\cdot, \cdot; \varepsilon)$ defined in (A.14) with $\mu(\cdot) = \mu_{k,\varepsilon}(\cdot)$ and $\nu(\cdot) = \nu_{k,\varepsilon}(\cdot)$. Consequently, the desired conclusion is true for the case that $k = \delta = \lambda + 1/2 \in (1/2, 1)$.

Finally, by virtue of the definition of k_ε, we have for each $\varepsilon > 0$ and $1/2 < \delta < \lambda + 1/2 = k < 1$,

$$\forall x \in \mathbb{R}_+, \quad x(\ln x)^\delta\mathbf{1}_{x>1} \leq \varepsilon(x+k_\varepsilon)\,(\ln(x+k_\varepsilon))^{\lambda+\frac{1}{2}}$$

and for each $\varepsilon > 0$ and $1/2 < \lambda + 1/2 < \delta = k < 1$,

$$\forall x \in \mathbb{R}_+, \quad (\ln(x+k_\varepsilon))^{2\lambda+1-\delta} \leq \varepsilon\,(\ln(x+k_\varepsilon))^{\delta}.$$

With the preceding two inequalities, in view of the definition of $\mu_{k,\varepsilon}(\cdot)$ with $k \in (1/2, 1)$, by a similar computation as above we can deduce that in both $1/2 < \delta < \lambda + 1/2 = k < 1$ and $1/2 < \lambda + 1/2 < \delta = k < 1$ cases, the desired conclusions are also true.

(iii) Case $k = 1$. We first consider the special case of $k = \delta = 2\lambda = 1$. Let $\varepsilon > 0$ and choose the following function

$$\bar{\phi}(s, x; \varepsilon) := (x + k_\varepsilon) \exp\left(\mu_s \ln(x + k_\varepsilon) + \nu_s\right) = e^{\nu_s}(x + k_\varepsilon)^{1+\mu_s}, \quad (s, x) \in [0, T] \times \mathbb{R}_+ \tag{A.24}$$

to explicitly solve the inequality (3.6) with $\delta = 1$ and $\lambda = 1/2$, where $\mu_s, \nu_s : [0, T] \to \mathbb{R}_+$ be two increasing and continuously differential functions with $\mu_0 = \varepsilon$ and $\nu_0 = 0$. For each $(s, x) \in [0, T] \times \mathbb{R}_+$, a simple computation gives

$$\bar{\phi}_x(s, x; \varepsilon) = \bar{\phi}(s, x; \varepsilon) \frac{1 + \mu_s}{x + k_\varepsilon} > 0, \tag{A.25}$$

$$\bar{\phi}_{xx}(s, x; \varepsilon) = \bar{\phi}(s, x; \varepsilon) \frac{\mu_s(1 + \mu_s)}{(x + k_\varepsilon)^2} > 0 \tag{A.26}$$

and

$$\bar{\phi}_s(s, x; \varepsilon) = \bar{\phi}(s, x; \varepsilon) \left(\mu_s' \ln(x + k_\varepsilon) + \nu_s'\right). \tag{A.27}$$

Note that

$$\forall x \in \mathbb{R}_+, \quad x \ln x \mathbf{1}_{x>1} \le (x + k_\varepsilon) \ln(x + k_\varepsilon).$$

We substitute (A.25), (A.26) and (A.27) into the left side of (3.6) with $\delta = 1$, $\lambda = 1/2$ and $\bar{\phi}(\cdot, \cdot; \varepsilon)$ instead of $\phi(\cdot, \cdot)$ to get that for each $(s, x) \in [0, T] \times \mathbb{R}_+$,

$$\begin{aligned}
& -\beta \bar{\phi}_x(s, x; \varepsilon) x \ln x \mathbf{1}_{x>1} - \frac{\gamma^2}{2} \frac{\left(\bar{\phi}_x(s, x; \varepsilon)\right)^2}{\bar{\phi}_{xx}(s, x; \varepsilon)} \left[(1 + \varepsilon) \left|\ln \frac{\gamma \bar{\phi}_x(s, x; \varepsilon)}{\bar{\phi}_{xx}(s, x; \varepsilon)}\right| + C_{\varepsilon,\lambda}\right] \\
& + \bar{\phi}_s(s, x; \varepsilon) \\
\ge\ & -\beta \bar{\phi}(s, x; \varepsilon)(1 + \mu_s) \ln(x + k_\varepsilon) - \frac{\gamma^2(1 + \mu_s)}{2\mu_s} \bar{\phi}(s, x; \varepsilon) \left[(1 + \varepsilon) \left|\ln \frac{\gamma(x + k_\varepsilon)}{\mu_s}\right| + C_{\varepsilon,\lambda}\right] \\
& + \bar{\phi}(s, x; \varepsilon) \left(\mu_s' \ln(x + k_\varepsilon) + \nu_s'\right) \\
\ge\ & \bar{\phi}(s, x; \varepsilon) \left[\left(-\beta(1 + \mu_s) - \frac{\gamma^2(1 + \mu_s)(1 + \varepsilon)}{2\mu_s} + \mu_s'\right) \ln(x + k_\varepsilon) \right. \\
& \left. + \left(-\frac{\gamma^2}{2}(1 + \mu_s)\bar{C}_\varepsilon + \nu_s'\right)\right],
\end{aligned}$$

where

$$\bar{C}_\varepsilon := \frac{(1 + \varepsilon)|\ln \gamma - \ln \varepsilon| + C_{\varepsilon,\lambda}}{\varepsilon}.$$

Thus, (3.6) with $\delta = 1$, $\lambda = 1/2$ and $\bar{\phi}(\cdot, \cdot; \varepsilon)$ instead of $\phi(\cdot, \cdot)$ holds if the functions $\mu_s, \nu_s \in [0, T]$ satisfies

$$\mu_s' = \beta \mu_s + \frac{\gamma^2(1 + \varepsilon)}{2\mu_s} + \frac{\gamma^2(1 + \varepsilon)}{2} + \beta, \quad s \in [0, T] \tag{A.28}$$

and

$$\nu_s = \frac{\gamma^2}{2} \bar{C}_\varepsilon \int_0^s (1 + \mu_r) \mathrm{d}r, \quad s \in [0, T]. \tag{A.29}$$

It is clear that $\mu_{k,\varepsilon}(\cdot)$ with $k = \delta = 2\lambda = 1$ is just the unique strictly increasing and continuous solution of ODE (A.28) with ε at the origin. And, we let, in view of (A.29),

$$\nu_{k,\varepsilon}(s) := \frac{\gamma^2}{2} \bar{C}_\varepsilon \int_0^s (1 + \mu_{k,\varepsilon}(r)) \mathrm{d}r, \quad s \in [0, T]. \tag{A.30}$$

Note that when $k=\delta=2\lambda=1$, the function $\varphi(\cdot,\cdot;k,\varepsilon)$ defined in (3.7) is just the function $\bar{\phi}(\cdot,\cdot;\varepsilon)$ defined in (A.24) with $\mu(\cdot)=\mu_{k,\varepsilon}(\cdot)$ and $\nu(\cdot)=\nu_{k,\varepsilon}(\cdot)$. Consequently, the desired conclusion is true for the case that $k=\delta=2\lambda=1$.

Finally, by virtue of the definition of k_ε, we have for each $\varepsilon>0$ and $0\le\delta<2\lambda=k=1$,

$$\forall x\in\mathbb{R}_+,\quad x(\ln x)^{\delta}\mathbf{1}_{x>1}\le\varepsilon(x+k_\varepsilon)\ln(x+k_\varepsilon)$$

and for each $\varepsilon>0$ and $0\le 2\lambda<\delta=k=1$,

$$\forall x\in\mathbb{R}_+,\quad (\ln(x+k_\varepsilon))^{2\lambda}\le\varepsilon\ln(x+k_\varepsilon).$$

With the preceding two inequalities, in view of the definition of $\mu_{k,\varepsilon}(\cdot)$ with $k=1$, by a similar computation as above we can deduce that in both $0\le\delta<2\lambda=k=1$ and $0\le 2\lambda<\delta=k=1$ cases, the desired conclusions are also true.

(iv) Case $k>1$. We first consider the special case of $k=2\lambda>1$ and $\delta=1$. Let $\varepsilon>0$ and choose the following function

$$\tilde{\phi}(s,x;\varepsilon):=(x+k_\varepsilon)\exp\left(\mu_s\left(\ln(x+k_\varepsilon)\right)^{2\lambda}+\nu_s\right),\quad (s,x)\in[0,T]\times\mathbb{R}_+ \tag{A.31}$$

to explicitly solve the inequality (3.6) with $\delta=1$, $\mu_s,\nu_s:[0,T]\to\mathbb{R}_+$ be two increasing and continuously differential functions with $\mu_0=\varepsilon$, $\mu_T=\mu_{k,\varepsilon}(T)$ and $\nu_0=1$. For each $(s,x)\in[0,T]\times\mathbb{R}_+$, a simple computation gives

$$\tilde{\phi}_x(s,x;\varepsilon)=\tilde{\phi}(s,x;\varepsilon)\frac{2\lambda\mu_s\left(\ln(x+k_\varepsilon)\right)^{2\lambda-1}+1}{x+k_\varepsilon}>0, \tag{A.32}$$

$$\begin{aligned}&\tilde{\phi}_{xx}(s,x;\varepsilon)\\&\quad=\tilde{\phi}(s,x;\varepsilon)\frac{2\lambda\mu_s\left(\ln(x+k_\varepsilon)\right)^{2\lambda-1}\left[\ln(x+k_\varepsilon)+2\lambda\mu_s\left(\ln(x+k_\varepsilon)\right)^{2\lambda}+(2\lambda-1)\right]}{(x+k_\varepsilon)^2\ln(x+k_\varepsilon)}>0\end{aligned} \tag{A.33}$$

and

$$\phi_s(s,x)=\tilde{\phi}(s,x;\varepsilon)\left(\mu_s'\left(\ln(x+k_\varepsilon)\right)^{2\lambda}+\nu_s'\right)>0. \tag{A.34}$$

Furthermore, for each $(s,x)\in[0,T]\times\mathbb{R}_+$, by (A.32), (A.33) and the definition of k_ε we have

$$\begin{aligned}\frac{\gamma^2}{2}\frac{\left(\tilde{\phi}_x(s,x;\varepsilon)\right)^2}{\tilde{\phi}_{xx}(s,x;\varepsilon)}&\le\frac{\gamma^2}{2}\tilde{\phi}(s,x;\varepsilon)\left(1+\frac{1}{2\lambda\mu_s\left(\ln(x+k_\varepsilon)\right)^{2\lambda-1}}\right)\\&\le\frac{\gamma^2}{2}\tilde{\phi}(s,x;\varepsilon)\left(1+\frac{\varepsilon}{2\lambda\mu_s}\right).\end{aligned} \tag{A.35}$$

And, from the definition of k_ε it can be directly verified that for each $(s,x)\in[0,T]\times\mathbb{R}_+$,

$$\ln(x+k_\varepsilon)+2\lambda\mu_s\left(\ln(x+k_\varepsilon)\right)^{2\lambda}+(2\lambda-1)\le\frac{1+\varepsilon}{\varepsilon}\left[\ln(x+k_\varepsilon)+2\lambda\mu_s\left(\ln(x+k_\varepsilon)\right)^{2\lambda}\right]. \tag{A.36}$$

In view of (A.32), (A.33) and (A.36) together with the definition of k_ε, we have

$$\left\{\begin{aligned}&\frac{\gamma\tilde{\phi}_x(s,x;\varepsilon)}{\tilde{\phi}_{xx}(s,x;\varepsilon)}\ge\frac{\gamma\varepsilon}{2\lambda(1+\varepsilon)\mu_s}\frac{x+k_\varepsilon}{\left(\ln(x+k_\varepsilon)\right)^{2\lambda-1}}\ge\frac{\gamma}{2\lambda\mu_T}\sqrt{x+k_\varepsilon}\ge\frac{\gamma}{2\lambda\mu_{k,\varepsilon}(T)}\sqrt{k_\varepsilon}\ge 1;\\&\frac{\gamma\tilde{\phi}_x(s,x;\varepsilon)}{\tilde{\phi}_{xx}(s,x;\varepsilon)}\le\frac{\gamma}{2\lambda\mu_s}\frac{x+k_\varepsilon}{\left(\ln(x+k_\varepsilon)\right)^{2\lambda-1}}\le\frac{\gamma}{2\lambda\varepsilon}(x+k_\varepsilon),\end{aligned}\right.$$

which yields

$$\left|\ln \frac{\gamma \tilde{\phi}_x(s,x;\varepsilon)}{\tilde{\phi}_{xx}(s,x;\varepsilon)}\right|^{2\lambda} \le 2^{2\lambda-1}\left|\ln \frac{\gamma}{2\lambda\varepsilon}\right|^{2\lambda} + 2^{2\lambda-1}\left(\ln(x+k_\varepsilon)\right)^{2\lambda}. \tag{A.37}$$

In the sequel, observe that

$$x\ln x\mathbf{1}_{x>1} \le (x+k_\varepsilon)\ln(x+k_\varepsilon), \quad x\in\mathbb{R}_+.$$

We substitute (A.35), (A.37), (A.32) and (A.34) into the left side of (3.6) with $\delta = 1$ and $\tilde{\phi}(\cdot,\cdot;\varepsilon)$ instead of $\phi(\cdot,\cdot)$ to get

$$\begin{aligned}
&-\beta\tilde{\phi}_x(s,x;\varepsilon)x\ln x\mathbf{1}_{x>1}\\
&-\frac{\gamma^2}{2}\frac{\left(\tilde{\phi}_x(s,x;\varepsilon)\right)^2}{\tilde{\phi}_{xx}(s,x;\varepsilon)}\left[4^{(\lambda-1)^+}(1+\varepsilon)\left|\ln\frac{\gamma\tilde{\phi}_x(s,x;\varepsilon)}{\tilde{\phi}_{xx}(s,x;\varepsilon)}\right|^{2\lambda}+C_{\varepsilon,\lambda}\right]+\tilde{\phi}_s(s,x;\varepsilon)\\
\ge{}& -\beta\tilde{\phi}(s,x;\varepsilon)\left[2\lambda\mu_s\left(\ln(x+k_\varepsilon)\right)^{2\lambda}+\ln(x+k_\varepsilon)\right]\\
&-\frac{\gamma^2}{2}\tilde{\phi}(s,x;\varepsilon)\left(1+\frac{\varepsilon}{2\lambda\mu_s}\right)\left[(1+\varepsilon)\bar{k}_\lambda\left(\ln(x+k_\varepsilon)\right)^{2\lambda}+\bar{C}_{\gamma,\lambda,\varepsilon}\right]\\
&+\tilde{\phi}(s,x;\varepsilon)\left(\mu_s'\left(\ln(x+k_\varepsilon)\right)^{2\lambda}+\nu_s'\right), \qquad (s,x)\in[0,T]\times\mathbb{R}_+,
\end{aligned}$$

where $\bar{k}_\lambda$ is defined in (2.15), and

$$\bar{C}_{\gamma,\lambda,\varepsilon} := (1+\varepsilon)\bar{k}_\lambda\left|\ln\frac{\gamma}{2\lambda\varepsilon}\right|^{2\lambda}+C_{\varepsilon,\lambda}.$$

Furthermore, in view of $k=2\lambda>1$ and the definition of k_ε, we know that

$$\forall x\in\mathbb{R}_+, \quad \ln(x+k_\varepsilon)\le\varepsilon\left(\ln(x+k_\varepsilon)\right)^{2\lambda}.$$

Then, for each $(s,x)\in[0,T]\times\mathbb{R}_+$, we have

$$\begin{aligned}
&-\beta\tilde{\phi}_x(s,x;\varepsilon)x\ln x\mathbf{1}_{x>1}-\frac{\gamma^2}{2}\frac{\left(\tilde{\phi}_x(s,x;\varepsilon)\right)^2}{\tilde{\phi}_{xx}(s,x;\varepsilon)}\left[4^{(\lambda-1)^+}(1+\varepsilon)\left|\ln\frac{\gamma\tilde{\phi}_x(s,x;\varepsilon)}{\tilde{\phi}_{xx}(s,x;\varepsilon)}\right|^{2\lambda}+C_{\varepsilon,\lambda}\right]\\
&+\tilde{\phi}_s(s,x;\varepsilon)\\
\ge{}&\tilde{\phi}(s,x;\varepsilon)\left\{\left[-\frac{\gamma^2(1+\varepsilon)\bar{k}_\lambda}{2}\left(1+\frac{\varepsilon}{2\lambda\mu_s}\right)-2\beta\lambda\mu_s-\beta\varepsilon+\mu_s'\right]\left(\ln(x+k_\varepsilon)\right)^{2\lambda}\right.\\
&\left.+\left[-\frac{\gamma^2}{2}\left(1+\frac{\varepsilon}{2\lambda\mu_s}\right)\bar{C}_{\gamma,\lambda,\varepsilon}+\nu_s'\right]\right\}.
\end{aligned}$$

Thus, (3.6) with $\delta=1$ and $\tilde{\phi}(\cdot,\cdot;\varepsilon)$ instead of $\phi(\cdot,\cdot)$ holds if the functions $\mu_s,\nu_s\in[0,T]$ satisfies

$$\mu_s' = \frac{\gamma^2(1+\varepsilon)\bar{k}_\lambda}{2}\left(1+\frac{\varepsilon}{2\lambda\mu_s}\right)+2\lambda\beta\mu_s+\beta\varepsilon \quad s\in[0,T] \tag{A.38}$$

and

$$\nu_s = \frac{\gamma^2\varepsilon}{4\lambda}\bar{C}_{\gamma,\lambda,\varepsilon}\int_0^s\frac{1}{\mu_r}\mathrm{d}r+\frac{\gamma^2}{2}\bar{C}_{\gamma,\lambda,\varepsilon}s, \quad s\in[0,T]. \tag{A.39}$$

It is clear that $\mu_{k,\varepsilon}(\cdot)$ with $k=2\lambda>1$ and $\delta=1$ is just the unique strictly increasing and continuous solution of ODE (A.38) with ε at the origin. And, we let, in view of (A.39),

$$\nu_{k,\varepsilon}(s) := \frac{\gamma^2\varepsilon}{4\lambda}\bar{C}_{\gamma,\lambda,\varepsilon}\int_0^s\frac{1}{\mu_{k,\varepsilon}(r)}\mathrm{d}r+\frac{\gamma^2}{2}\bar{C}_{\gamma,\lambda,\varepsilon}s, \quad s\in[0,T]. \tag{A.40}$$

Note that when $k = 2\lambda > 1$ and $\delta = 1$, the function $\varphi(\cdot, \cdot; k, \varepsilon)$ defined in (3.7) is just the function $\tilde{\phi}(\cdot, \cdot; \varepsilon)$ defined in (A.31) with $\mu(\cdot) = \mu_{k,\varepsilon}(\cdot)$ and $\nu(\cdot) = \nu_{k,\varepsilon}(\cdot)$. Consequently, the desired conclusion is true for the case that $k = 2\lambda > 1$ and $\delta = 1$.

Finally, by virtue of the definition of k_ε, we have for each $\varepsilon > 0$, $k = 2\lambda > 1$ and $0 \le \delta < 1$,

$$\forall x \in \mathbb{R}_+, \quad x(\ln x)^\delta \mathbf{1}_{x>1} \le \varepsilon(x + k_\varepsilon)\ln(x + k_\varepsilon).$$

With the last inequality, in view of the definition of $\mu_{k,\varepsilon}(\cdot)$ with $k = 2\lambda > 1$, by a similar computation as above we can deduce that in the case that $k > 1$ and $0 \le \delta < 1$, the desired conclusion is also true.

On the whole, the proof of Proposition 3.4 is then complete. $\square$

Proof of Proposition 3.5. The first inequality in (3.8) is obvious. We now prove the second inequality. In fact, for each $\varepsilon \ge 0$ we have

$$\begin{aligned}
\frac{\varphi(s, x; k, \varepsilon)}{\psi(x, \mu_{k,\varepsilon}(s); k) + 1} &= \frac{(x + k_\varepsilon)\exp\left(\mu_{k,\varepsilon}(s)\,(\ln(x + k_\varepsilon))^k + \nu_{k,\varepsilon}(s)\right)}{x\exp\left(\mu_{k,\varepsilon}(s)\,(\ln(1 + x))^k\right) + 1} \\
&\le \frac{x + k_\varepsilon}{x}\exp\left(\mu_{k,\varepsilon}(T)\left((\ln(x + k_\varepsilon))^k - (\ln(1 + x))^k\right) + \nu_{k,\varepsilon}(T)\right) \\
&=: H_1(x; k, \beta, \gamma, \lambda, \varepsilon, T), \qquad (s, x) \in [0, T] \times [1, +\infty)
\end{aligned}$$

with k_ε and $\nu_{k,\varepsilon}(\cdot)$ being defined in Proposition 3.4. And, in the case of $x \in [0, 1]$,

$$\begin{aligned}
\frac{\varphi(s, x; k, \varepsilon)}{\psi(x, \mu_{k,\varepsilon}(s); k) + 1} &\le (1 + k_\varepsilon)\exp\left(\mu_{k,\varepsilon}(T)\,(\ln(1 + k_\varepsilon))^k + \nu_{k,\varepsilon}(T)\right) \\
&=: H_2(k, \beta, \gamma, \lambda, \varepsilon, T), \qquad s \in [0, T].
\end{aligned}$$

Hence, for all $x \in \mathbb{R}_+$, we have

$$\frac{\varphi(s, x; k, \varepsilon)}{\psi(x, \mu_{k,\varepsilon}(s); k) + 1} \le H_1(x; k, \beta, \gamma, \lambda, \varepsilon, T)\mathbf{1}_{x\ge 1} + H_2(k, \beta, \gamma, \lambda, \varepsilon, T)\mathbf{1}_{0\le x<1}, \quad s \in [0, T]. \tag{A.41}$$

Thus, in view of (A.41) and the fact that the function $H_1(x; k, \beta, \gamma, \lambda, \varepsilon, T)$ is continuous on $[1, +\infty)$ and tends to $\exp(\nu_{k,\varepsilon}(T))$ as $x \to +\infty$, the second inequality in (3.8) follows immediately. $\square$

Proof of Proposition 3.8. Let $q \in (0, p)$, $0 < \epsilon \le p/q - 1$, and pick a sufficiently large constant $\tilde{k}_\epsilon > 1$ depending only on ϵ and λ such that

$$\forall x \in \mathbb{R}_+, \quad \left(\ln(x + \tilde{k}_\epsilon)\right)^{2\lambda+1} \le (x + \tilde{k}_\epsilon)^{2\epsilon}. \tag{A.42}$$

For any fixed $\lambda > 0$, with $y/2^{(\lambda-1)^+ + 2}$ instead of the variable y in Proposition 3.2 we know the existence of a constant $C_\lambda > 0$ depending only on λ such that for each $x, y > 0$,

$$2xy\left|\ln y - ((\lambda - 1)^+ + 2)\ln 2\right|^\lambda \le 2^{(\lambda-1)^+ + 2}\left(4^{(\lambda-1)^+ + 1}|\ln x|^{2\lambda} + C_\lambda\right)x^2 + \frac{y^2}{2^{(\lambda-1)^+ + 2}}. \tag{A.43}$$

For each $x, y > 0$, we also have

$$|\ln y|^\lambda \le 2^{(\lambda-1)^+}\left|\ln y - ((\lambda - 1)^+ + 2)\ln 2\right|^\lambda + 2^{(\lambda-1)^+}|((\lambda - 1)^+ + 2)\ln 2|^\lambda \tag{A.44}$$

and

$$2xy\left(2^{(\lambda-1)^+}|((\lambda-1)^++2)\ln 2|^\lambda\right)\leq 4^{(\lambda-1)^++1}|((\lambda-1)^++2)\ln 2|^{2\lambda}x^2+y^2/4. \quad (A.45)$$

By multiplying $2xy$ into both sides of (A.44) and using (A.43) and (A.45), we have for a constant $K_\lambda>0$ depending only on λ and C_λ,

$$2xy|\ln y|^\lambda\leq K_\lambda\left(|\ln x|^{2\lambda}+1\right)x^2+y^2/2,\quad x,y>0,$$

and then, in view of (A.42), there exists a constant $\bar{K}_{\lambda,\gamma}>0$ depending only on (γ,K_λ) such that

$$\forall x>k_\varepsilon \text{ and } y>0,\quad 2\gamma xy|\ln y|^\lambda\leq \bar{K}_{\lambda,\gamma}x^2|\ln x|^{2\lambda}+y^2/2\leq \bar{K}_{\lambda,\gamma}x^{2+2\epsilon}+y^2/2. \quad (A.46)$$

We also remark that (A.46) still holds for $\lambda=0$ due to the basic inequality

$$\forall x,y\geq 0,\quad 2\gamma xy\leq 2\gamma^2x^2+y^2/2$$

Consequently, (A.46) holds for each $\lambda\geq 0$.

For each $n\geq 1$, define the following stopping time

$$\tau_n:=\inf\left\{s\in[0,T]:\int_0^s(|Y_r|+\tilde{k}_\epsilon)^2|Z_r|^2\mathrm{d}r\geq n\right\}\wedge T$$

with the convention that $\inf\emptyset=+\infty$. Applying Itô-Tanaka's formula to $(|Y_s|+\tilde{k}_\epsilon)^2$, we have

$$\begin{aligned}&(|Y_0|+\tilde{k}_\epsilon)^2+\int_0^{\tau_n}|Z_s|^2\mathrm{d}s\\ \leq&(|Y_{\tau_n}|+\tilde{k}_\epsilon)^2+2\int_0^{\tau_n}(|Y_s|+\tilde{k}_\epsilon)\mathrm{sgn}(Y_s)g(s,Y_s,Z_s)\mathrm{d}s\\ &+2\int_0^{\tau_n}(|Y_s|+\tilde{k}_\epsilon)\mathrm{sgn}(Y_s)Z_s\cdot\mathrm{d}B_s. \end{aligned}\quad (A.47)$$

From (EX2) together with (A.42) and (A.46), we have

$$\begin{aligned}&2(|Y_s|+\tilde{k}_\epsilon)\mathrm{sgn}(Y_s)g(s,Y_s,Z_s)\\ \leq&2\alpha_s(|Y_s|+\tilde{k}_\epsilon)+2\beta(|Y_s|+\tilde{k}_\epsilon)^2\left(\ln(|Y_s|+\tilde{k}_\epsilon)\right)^\delta+2\gamma(|Y_s|+\tilde{k}_\epsilon)|Z_s\|\ln|Z_s\|^\lambda\\ \leq&2\alpha_s(|Y_s|+\tilde{k}_\epsilon)+(2\beta+\bar{K}_{\lambda,\gamma})(|Y_s|+\tilde{k}_\epsilon)^{2+2\epsilon}+|Z_s|^2/2.\end{aligned}$$

Then, coming back to (A.47), we have for each $n\geq 1$,

$$\begin{aligned}\int_0^{\tau_n}|Z_s|^2\mathrm{d}s\leq&\ 2\sup_{s\in[0,T]}(|Y_s|+\tilde{k}_\epsilon)^2+4\sup_{s\in[0,T]}(|Y_s|+\tilde{k}_\epsilon)\int_0^T\alpha_s\mathrm{d}s\\ &+4(\beta+\bar{K}_{\lambda,\gamma})\sup_{s\in[0,T]}(|Y_s|+\tilde{k}_\epsilon)^{2+2\epsilon}+4\int_0^{\tau_n}(|Y_s|+\tilde{k}_\epsilon)\mathrm{sgn}(Y_s)Z_s\cdot\mathrm{d}B_s,\end{aligned}$$

and then for a positive constant $c_q > 0$ depending only on $(q, \beta, \bar{K}_{\lambda,\gamma})$,

$$\begin{aligned}
&\mathbb{E}\left[\left(\int_0^{\tau_n} |Z_s|^2 \mathrm{d}s\right)^{\frac{q}{2}}\right] \\
\leq\ & c_q\,\mathbb{E}\left[\sup_{s\in[0,T]} (|Y_s| + \tilde{k}_\epsilon)^q\right] + c_q\,\mathbb{E}\left[\left(\sup_{s\in[0,T]} (|Y_s| + \tilde{k}_\epsilon) \int_0^T \alpha_s \mathrm{d}s\right)^{\frac{q}{2}}\right] \\
&+ c_q\,\mathbb{E}\left[\sup_{s\in[0,T]} (|Y_s| + \tilde{k}_\epsilon)^{(1+\epsilon)q}\right] + c_q\,\mathbb{E}\left[\left(\int_0^{\tau_n} (|Y_s| + \tilde{k}_\epsilon)\mathrm{sgn}(Y_s) Z_s \cdot \mathrm{d}B_s\right)^{\frac{q}{2}}\right].
\end{aligned} \tag{A.48}$$

Moreover, using the BDG inequality and Young's inequality, we have for a constant $d_q > 0$ depending only on c_q,

$$\begin{aligned}
&c_q \mathbb{E}\left[\left(\int_0^{\tau_n} (|Y_s| + \tilde{k}_\epsilon)\mathrm{sgn}(Y_s) Z_s \cdot \mathrm{d}B_s\right)^{\frac{q}{2}}\right] \\
&\quad\leq d_q\,\mathbb{E}\left[\sup_{s\in[0,T]} (|Y_s| + \tilde{k}_\epsilon)^q\right] + \frac{1}{2}\mathbb{E}\left[\left(\int_0^{\tau_n} |Z_s|^2 \mathrm{d}s\right)^{\frac{q}{2}}\right].
\end{aligned}$$

Substituting the last inequality into (A.48) and using Hölder's inequality, we have for each $n \geq 1$,

$$\begin{aligned}
\frac{1}{2}\mathbb{E}\left[\left(\int_0^{\tau_n} |Z_s|^2 \mathrm{d}s\right)^{\frac{q}{2}}\right] \leq\ & (c_q + d_q)\mathbb{E}\left[\sup_{s\in[0,T]} (|Y_s| + \tilde{k}_\epsilon)^q\right] \\
&+ c_q\,\mathbb{E}\left[\sup_{s\in[0,T]} (|Y_s| + \tilde{k}_\epsilon)^{(1+\epsilon)q}\right] \\
&+ c_q \left(\mathbb{E}\left[\sup_{s\in[0,T]} (|Y_s| + \tilde{k}_\epsilon)^q\right]\right)^{\frac{1}{2}} \left(\mathbb{E}\left[\left(\int_0^T \alpha_s \mathrm{d}s\right)^q\right]\right)^{\frac{1}{2}}.
\end{aligned}$$

Since $|Y_\cdot| + \int_0^\cdot \alpha_s \mathrm{d}s \in \mathcal{S}^p$ and $q < (1+\epsilon)q < p$, using Fatou's Lemma, we have $Z_\cdot \in \mathcal{M}^q$. The proof is then complete. $\square$

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